# COF and wear of 304 stainless steel in dry reciprocating sliding

## ME 323, IIT Bombay - final report (generated by run_all.py)

23 runs, ball-on-flat (Ø 6 mm bearing-steel ball), A = 5 mm, L 10–100 N, f 4–15 Hz, t 60–120 s. Generated 2026-10-27 17:30.

| target | final model | selected ML model | $R^2$ P1 | $R^2$ P2 |
|---|---|---|---|---|
| COF, first 10 s (primary) | $\mu_0 + c \cdot L^{(-1/3)}$ | ANN + physics | +0.36 | +0.23 |
| COF, end of run (second-half median) | $\mu_0 + c \cdot L^{(-1/3)}$ | none | +0.08 | +0.05 |
| log10 wear volume V [$mm^3$] | Archard (slope 1) | ANN | +0.29 | +0.15 |
| log10 track width [µm] | a + b·ln L | GPR + physics | +0.20 | +0.39 |
| log10 k [$mm^3/(N \cdot m)$] | Archard constant k | ANN + physics | -0.15 | -0.20 |

# Data summary

| run_id | role | replicate_group | test_date | Load_N | Freq_Hz | Run_time_s | cycles | S_m | COF_first10s | COF_end | wear_status | wear_location | k_mm3_Nm | k_upper_mm3_Nm | track_width_um |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| R01 | training | R01 | 2026-10-13 | 10 | 6 | 120 | 719 | 14.4 | 0.141 | 0.145 | detected | a | 3.88e-05 | – | 442 |
| R02 | training | R02 | 2026-10-13 | 15 | 4 | 120 | 479 | 9.59 | 0.159 | 0.165 | detected | a | 1.88e-05 | – | 311 |
| R04 | training | R04 | 2026-10-13 | 32 | 7.5 | 84 | 628 | 12.6 | 0.135 | 0.152 | below_LOD | – | – | 6.64e-06 | 424 |
| R05 | training | R05 | 2026-10-13 | 10 | 15 | 60 | 896 | 17.9 | 0.159 | 0.153 | detected | c | 1.13e-05 | – | 386 |
| R06 | training | R06 | 2026-10-13 | 24.5 | 4.5 | 116 | 521 | 10.4 | 0.135 | 0.149 | below_LOD | – | – | 1.07e-05 | 349 |
| R07 | training | R07 | 2026-10-13 | 10.5 | 12.5 | 68 | 847 | 16.9 | 0.129 | 0.139 | below_LOD | – | – | 1.02e-05 | 238 |
| R08 | training | R08 | 2026-10-13 | 54 | 12 | 72 | 861 | 17.2 | 0.127 | 0.146 | detected | c | 5.22e-06 | – | 491 |
| R09 | training | R09 | 2026-10-13 | 20 | 11 | 62 | 679 | 13.6 | 0.148 | 0.16 | below_LOD | – | – | 1.07e-05 | 335 |
| R10 | training | R10 | 2026-10-13 | 40.5 | 6.5 | 72 | 467 | 9.33 | 0.131 | 0.139 | below_LOD | – | – | 7.86e-06 | 361 |
| R11 | training | R11 | 2026-10-16 | 100 | 4 | 60 | 238 | 4.77 | 0.13 | 0.14 | detected | c | 1.34e-04 | – | 642 |
| R12 | training | R12 | 2026-10-16 | 100 | 7.5 | 120 | 897 | 17.9 | 0.119 | 0.118 | not_measurable | – | – | – | 733 |
| R13 | training | R13 | 2026-10-16 | 16 | 5.5 | 82 | 449 | 8.98 | 0.126 | 0.128 | detected | b | 5.84e-05 | – | 266 |
| R14 | training | R04 | 2026-10-16 | 32 | 7.5 | 84 | 627 | 12.5 | 0.125 | 0.142 | detected | a | 6.11e-05 | – | 370 |
| R17 | training | R17 | 2026-10-16 | 15 | 6.5 | 100 | 648 | 12.9 | 0.129 | 0.132 | detected | c | 6.11e-05 | – | 305 |
| R19 | training | R19 | 2026-10-16 | 89 | 7.5 | 68 | 507 | 10.1 | 0.118 | 0.121 | detected | b | 4.46e-05 | – | 618 |
| R20 | training | R04 | 2026-10-16 | 32 | 7.5 | 84 | 628 | 12.6 | 0.126 | 0.14 | detected | b | 4.92e-05 | – | 318 |
| R21 | training | R21 | 2026-10-16 | 47.5 | 7 | 108 | 753 | 15.1 | 0.12 | 0.126 | detected | m | 1.78e-05 | – | 321 |
| R22 | training | R22 | 2026-10-16 | 80 | 4 | 70 | 278 | 5.57 | 0.129 | 0.129 | detected | b | 6.53e-05 | – | 458 |
| R24 | test | R24 | 2026-10-16 | 23 | 5 | 64 | 318 | 6.36 | 0.131 | 0.131 | detected | c | 9.80e-06 | – | 133 |
| R25 | test | R25 | 2026-10-16 | 12 | 9 | 86 | 770 | 15.4 | 0.136 | 0.131 | detected | m | 2.51e-05 | – | 187 |
| R26 | test | R26 | 2026-10-16 | 73 | 8 | 102 | 813 | 16.3 | 0.117 | 0.129 | detected | a | 1.49e-05 | – | 421 |
| R28 | test | R28 | 2026-10-16 | 64 | 6 | 64 | 382 | 7.64 | 0.121 | 0.13 | detected | m | 2.89e-05 | – | 227 |
| R29 | test | R29 | 2026-10-16 | 13.5 | 5 | 108 | 538 | 10.8 | 0.15 | 0.151 | below_LOD | – | – | 2.41e-05 | 264 |

Missing DOE runs (no folder): R00, R03, R15, R16, R18, R23, R27. Wear: 16 detected, 6 below LOD, 1 not measurable.

# Final model cards

## Model card: COF, first 10 s (primary)

**Final model:** $\mu_0$ + c·L^(–1/3)  **Selected ML model:** ANN + physics  **Equation:** COF first 10 s = $\mu_0$ + c·L^(–1/3)

**Selection rule:** ANN + physics is the selected ML model (beats the constant in P1 and P2, tied with the best P1 RMSE, hybrid preferred) but does not beat the physics baseline robustly (lower RMSE and not 1-SE-tied in both P1 and P2).

**Noise floor:** replicate SD 0.0056 (R04,R14,R20); spread over runs / replicate SD = 2.2; n = 23 runs.

| param | estimate | ci_lo | ci_hi |
|---|---|---|---|
| mu0 | 0.0995 | 0.0864 | 0.114 |
| c | 0.0978 | 0.0492 | 0.142 |

| model | R² P1 | RMSE P1 | R² P2 | RMSE P2 | R² P3 | RMSE P3 | cov95 P1 % | perm p |
|---|---|---|---|---|---|---|---|---|
| $\mu_0$ + c·L^(–1/3) | 0.36 | 0.00947 | 0.225 | 0.0104 | 0.702 | 0.00638 | 95.7 | 5.00e-04 |
| ANN + physics | 0.43 | 0.00893 | 0.0385 | 0.0116 | 0.813 | 0.00506 | 17.4 | 0.0099 |
| Constant | -0.0939 | 0.0124 | -0.124 | 0.0126 | -0.0196 | 0.0118 | 91.3 | – |

## Model card: COF, end of run (second-half median)

**Final model:** $\mu_0$ + c·L^(–1/3)  **Selected ML model:** none  **Equation:** COF end = $\mu_0$ + c·L^(–1/3)

**Selection rule:** required ML models beat the constant in P1 and P2, but none is tied with the best P1 RMSE.

**Noise floor:** replicate SD 0.0062 (R04,R14,R20); spread over runs / replicate SD = 2.0; n = 23 runs.

| param | estimate | ci_lo | ci_hi |
|---|---|---|---|
| mu0 | 0.116 | 0.1 | 0.134 |
| c | 0.069 | 0.0176 | 0.118 |

| model | R² P1 | RMSE P1 | R² P2 | RMSE P2 | R² P3 | RMSE P3 | cov95 P1 % | perm p |
|---|---|---|---|---|---|---|---|---|
| $\mu_0$ + c·L^(–1/3) | 0.0782 | 0.0115 | 0.052 | 0.0122 | -0.376 | 0.00973 | 95.7 | 0.0215 |
| Constant | -0.0981 | 0.0126 | -0.125 | 0.0133 | -0.487 | 0.0101 | 95.7 | – |

# Model card: log10 wear volume V [mm³]

**Final model:** Archard (slope 1)  **Selected ML model:** ANN  **Equation:** log10 V = log10 k + log10(L·S)

**Selection rule:** ANN is the selected ML model (beats the constant in P1 and P2, tied with the best P1 RMSE) but does not beat the physics baseline robustly (lower RMSE and not 1-SE-tied in both P1 and P2).

**Noise floor:** replicate SD 0.0668 (R14,R20); spread over runs / replicate SD = 7.1; n = 16 runs.

| param | estimate | ci_lo | ci_hi |
|---|---|---|---|
| log10_k | -4.53 | -4.73 | -4.36 |

| model | R² P1 | RMSE P1 | R² P2 | RMSE P2 | R² P3 | RMSE P3 | cov95 P1 % | perm p |
|---|---|---|---|---|---|---|---|---|
| Archard (slope 1) | 0.289 | 0.387 | 0.151 | 0.412 | 0.383 | 0.341 | 93.8 | 0.009 |
| ANN | 0.737 | 0.236 | 0.162 | 0.41 | -0.837 | 0.589 | 100 | – |
| Constant | -0.149 | 0.492 | -0.205 | 0.491 | -0.369 | 0.509 | 100 | – |

# Model card: log10 track width [µm]

**Final model:** a + b·ln L  **Selected ML model:** GPR + physics  **Equation:** log10 w = a + b·ln L

**Selection rule:** GPR + physics is the selected ML model (beats the constant in P1 and P2, tied with the best P1 RMSE, hybrid preferred) but does not beat the physics baseline robustly (lower RMSE and not 1-SE-tied in both P1 and P2).

**Noise floor:** replicate SD 0.0625 (R04,R14,R20); spread over runs / replicate SD = 2.7; n = 23 runs.

| param | estimate | ci_lo | ci_hi |
|---|---|---|---|
| a | 2.11 | 1.81 | 2.37 |
| b | 0.126 | 0.0529 | 0.204 |
| exponent_beta (w ~ L^beta) | 0.291 | 0.122 | 0.469 |
| Hertz expectation for b | 0.145 | – | – |

| model | R² P1 | RMSE P1 | R² P2 | RMSE P2 | R² P3 | RMSE P3 | cov95 P1 % | perm p |
|---|---|---|---|---|---|---|---|---|
| a + b·ln L | 0.198 | 0.15 | 0.393 | 0.1 | -1.54 | 0.265 | 91.3 | 0.005 |
| GPR + physics | 0.262 | 0.144 | 0.548 | 0.0866 | -1.54 | 0.265 | 87 | 0.00997 |
| Constant | -0.0925 | 0.175 | -0.12 | 0.136 | -1.96 | 0.286 | 95.7 | – |

# Model card: log10 k [mm³/(N·m)]

**Final model:** Archard constant k  **Selected ML model:** ANN + physics  **Equation:** log10 k = constant (Archard)

**Selection rule:** ANN + physics beats the constant and, beyond the 1-SE tie, the physics baseline in both P1 and P2, and is tied with the best P1 RMSE; but without its most influential run (R11) it no longer beats the constant in P1, so the gain is not robust.

**k as one material constant:** 2.96e-05 mm³/(N·m) ×/÷ 2.36 (95 % bootstrap CI of the geometric mean 1.86e-05 – 4.38e-05).

**Noise floor:** replicate SD 0.0670 (R14,R20); spread over runs / replicate SD = 5.6; n = 16 runs.

| param | estimate | ci_lo | ci_hi |
|---|---|---|---|
| log10_k | -4.53 | -4.73 | -4.36 |

| model | R² P1 | RMSE P1 | R² P2 | RMSE P2 | R² P3 | RMSE P3 | cov95 P1 % | perm p |
|---|---|---|---|---|---|---|---|---|
| Archard constant k | -0.149 | 0.387 | -0.201 | 0.412 | -2.34 | 0.341 | 93.8 | – |
| ANN + physics | 0.399 | 0.28 | 0.386 | 0.295 | -4.47 | 0.437 | 87.5 | 0.0198 |

# Comparison studies C1–C6

## C1 – Which COF definition?

**Verdict:** Use COF_first10s: it follows the adhesion law best among the definitions (P1 $R^2$ +0.36, P2 +0.23; end-of-run COF +0.08) with signal-to-noise 2.2; the instrument mean reads 13 % low and its ratio to ours falls with frequency (r = -0.70).

| label | n | mean | replicate_sd | signal_to_noise | physics_R2_P1 | physics_R2_P2 | c_adhesion |
|---|---|---|---|---|---|---|---|
| first 10 s (primary) | 23 | 0.132 | 0.00557 | 2.17 | 0.36 | 0.225 | 0.0978 |
| 0.5-2 m of sliding | 23 | 0.132 | 0.00665 | 1.79 | 0.306 | 0.175 | 0.0912 |
| end: second-half median | 23 | 0.139 | 0.00619 | 1.98 | 0.0782 | 0.052 | 0.069 |
| last 10 s | 23 | 0.141 | 0.00603 | 2.13 | 0.0314 | -0.01 | 0.0674 |
| whole run (mean) | 23 | 0.137 | 0.00664 | 1.75 | 0.171 | 0.128 | 0.0746 |
| instrument \|Fx\|/Fz mean | 23 | 0.119 | 0.00577 | 1.55 | -0.138 | -0.195 | 0.0224 |

# C2 – Which wear measure?

**Verdict:** Use the most-worn valid location with robust levelling: 16 runs measurable with an LOD, replicate SD 0.07 dex, Archard $R^2$ P1 +0.29 / P2 +0.15; the old mean-of-locations, plain-levelling measure gives 0.34 dex and $R^2$ +0.08 / +0.01. Exponents: V ~ L^0.96 [0.41, 1.59] S^-0.07 [-1.32, 1.61] (Archard: 1, 1). The TA depth rule scores a higher $R^2$ (+0.52) only because it drops the shallow part of every groove: runs at <= 25 N keep a median 11 % of their groove, runs at >= 60 N 91 %, which inflates the load exponent to 1.9; it is not a volume.

| label | runs_measurable | replicate_sd_dex | signal_to_noise | best_model | best_R2_P1 | best_R2_P2 | exp_L | exp_L_ci | exp_S | exp_S_ci |
|---|---|---|---|---|---|---|---|---|---|---|
| V most-worn location, robust levelling (final) | 16 | 0.0668 | 7.1 | physics | 0.289 | 0.151 | 0.956 | [0.41, 1.59] | -0.067 | [-1.32, 1.61] |
| V mean of locations, robust levelling | 22 | 0.48 | 2.23 | physics | 0.122 | 0.0518 | 1.88 | [0.72, 3.18] | -0.0696 | [-1.91, 2.02] |
| V most-worn location, plain levelling | 13 | 0.102 | 2.95 | powerlaw | -0.0726 | -0.414 | 0.38 | [-0.06, 0.98] | -0.572 | [-1.21, 0.61] |
| V mean of locations, plain levelling (old) | 21 | 0.336 | 3.4 | physics | 0.0813 | 0.0131 | 1.91 | [0.43, 4.00] | 1.17 | [-0.95, 4.60] |
| V most-worn, 2-D box high-pass (anchor implementation) | 15 | 0.0621 | 6.73 | physics | 0.19 | -0.0737 | 0.811 | [0.27, 1.49] | 0.168 | [-1.15, 1.72] |
| V most-worn, TA depth rule | 15 | 0.00884 | 93.7 | powerlaw | 0.515 | 0.319 | 1.88 | [1.20, 2.71] | -0.228 | [-1.51, 1.60] |
| max wear depth | 21 | 0.504 | 1.04 | powerlaw | 0.0136 | -0.176 | – | – | – | – |
| track width | 23 | 0.0625 | 2.75 | physics | 0.198 | 0.393 | – | – | – | – |

# C3 – Which model?

**Verdict:** COF first 10 s: final = physics law (selected ML: ANN + physics); COF end: final = physics law (selected ML: none); log10 V: final = physics law (selected ML: ANN); log10 width: final = physics law (selected ML: GPR + physics); log10 k: final = physics law (selected ML: ANN + physics).

| target | label | P1_R2 | P1_RMSE | P2_R2 | P3_R2 | n_params | uncertainty | physically_consistent | perm_p | verdict |
|---|---|---|---|---|---|---|---|---|---|---|
| COF_first10s | Constant | -0.0939 | 0.0124 | -0.124 | -0.0196 | 1 | yes | no | – | reference |
| COF_first10s | Physics | 0.36 | 0.00947 | 0.225 | 0.702 | 2 | yes | yes | 5.00e-04 | physics baseline (final model) |
| COF_first10s | Power law (OLS) | 0.185 | 0.0107 | -0.0137 | 0.619 | 4 | yes | yes | – | worse than physics |
| COF_first10s | Ridge | 0.15 | 0.0109 | -0.107 | 0.626 | 4 | no | yes | – | worse than physics |
| COF_first10s | SVR | -0.207 | 0.013 | -0.116 | 0.69 | 23 SV | no | yes | – | no better than constant |
| COF_first10s | XGBoost | 0.0924 | 0.0113 | -0.208 | 0.446 | 200 trees, depth 1 | no | yes | – | unstable (beats constant in P1 only) |
| COF_first10s | ANN | 0.143 | 0.011 | 0.154 | 0.603 | 21 | yes | yes | – | worse than physics |
| COF_first10s | GPR | -0.169 | 0.0128 | -0.422 | 0.0734 | 5 | yes | yes | – | no better than constant |
| COF_first10s | TabPFN | – | – | – | – | – | – | – | – | not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download) |
| COF_first10s | ANN + physics | 0.43 | 0.00893 | 0.0385 | 0.813 | 23 | yes | yes | 0.0099 | chosen (selected ML model); not better than physics |
| COF_first10s | GPR + physics | 0.148 | 0.0109 | 0.0227 | 0.645 | 7 | yes | yes | 0.0399 | worse than physics |
| COF_first10s | TabPFN + physics | – | – | – | – | – | – | – | – | not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download) |
| COF_end | Constant | -0.0981 | 0.0126 | -0.125 | -0.487 | 1 | yes | yes | – | reference |
| COF_end | Physics | 0.0782 | 0.0115 | 0.052 | -0.376 | 2 | yes | yes | 0.0215 | physics baseline (final model) |
| COF_end | Power law (OLS) | -0.112 | 0.0127 | -0.146 | -0.842 | 4 | yes | yes | – | no better than constant |
| COF_end | Ridge | -0.22 | 0.0133 | -0.251 | -0.842 | 4 | no | yes | – | no better than constant |

| target | label | P1_R2 | P1_RMSE | P2_R2 | P3_R2 | n_params | uncertainty | physically_consistent | perm_p | verdict |
|---|---|---|---|---|---|---|---|---|---|---|
| COF_end | SVR | 0.608 | 0.00752 | 0.0358 | 0.493 | 21 SV | no | no | – | tied with best (not a required model) |
| COF_end | XGBoost | -0.128 | 0.0128 | -0.17 | -0.969 | 200 trees, depth 1 | no | yes | – | no better than constant |
| COF_end | ANN | 0.396 | 0.00933 | 0.00362 | 0.512 | 21 | yes | no | – | beats constant, not tied with best |
| COF_end | GPR | 0.374 | 0.0095 | -0.383 | -0.141 | 5 | yes | no | – | unstable (beats constant in P1 only) |
| COF_end | TabPFN | – | – | – | – | – | – | – | – | not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download) |
| COF_end | ANN + physics | 0.156 | 0.011 | -0.587 | 0.62 | 18 | yes | no | 0.158 | unstable (beats constant in P1 only) |
| COF_end | GPR + physics | 0.203 | 0.0107 | -0.224 | -0.531 | 7 | yes | yes | 0.0133 | unstable (beats constant in P1 only) |
| COF_end | TabPFN + physics | – | – | – | – | – | – | – | – | not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download) |
| log10_V | Constant | -0.149 | 0.492 | -0.205 | -0.369 | 1 | yes | no | – | reference |
| log10_V | Physics | 0.289 | 0.387 | 0.151 | 0.383 | 1 | yes | yes | 0.009 | physics baseline (final model) |
| log10_V | Power law (OLS) | 0.193 | 0.412 | 0.169 | -0.338 | 4 | yes | no | – | worse than physics |
| log10_V | Ridge | 0.159 | 0.421 | 0.178 | -0.358 | 4 | no | no | – | worse than physics |
| log10_V | SVR | 0.0657 | 0.444 | 0.529 | -0.571 | 16 SV | no | no | – | worse than physics |
| log10_V | XGBoost | 0.21 | 0.408 | 0.332 | 0.0351 | 200 trees, depth 2 | no | no | – | worse than physics |
| log10_V | ANN | 0.737 | 0.236 | 0.162 | -0.837 | 21 | yes | no | – | chosen (selected ML model); not better than physics |
| log10_V | GPR | 0.0196 | 0.454 | 0.795 | -1.51 | 5 | yes | no | – | worse than physics |

| target | label | P1_R2 | P1_RMSE | P2_R2 | P3_R2 | n_params | uncertainty | physically_consistent | perm_p | verdict |
|---|---|---|---|---|---|---|---|---|---|---|
| log10_V | TabPFN | – | – | – | – | – | – | – | – | not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download) |
| log10_V | ANN + physics | 0.629 | 0.28 | 0.549 | -0.031 | 22 | yes | no | 0.0099 | beats constant, not tied with best |
| log10_V | GPR + physics | 0.0816 | 0.44 | 0.216 | -0.413 | 6 | yes | no | 0.0498 | worse than physics |
| log10_V | TabPFN + physics | – | – | – | – | – | – | – | – | not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download) |
| log10_width | Constant | -0.0925 | 0.175 | -0.12 | -1.96 | 1 | yes | no | – | reference |
| log10_width | Physics | 0.198 | 0.15 | 0.393 | -1.54 | 2 | yes | yes | 0.005 | physics baseline (final model) |
| log10_width | Power law (OLS) | 0.0758 | 0.161 | 0.134 | -1.31 | 4 | yes | no | – | tied with best (not a required model) |
| log10_width | Ridge | 0.0078 | 0.167 | 0.0241 | -1.31 | 4 | no | yes | – | worse than physics |
| log10_width | SVR | 0.204 | 0.15 | 0.0817 | -1.3 | 22 SV | no | yes | – | tied with best (not a required model) |
| log10_width | XGBoost | 0.102 | 0.159 | 0.478 | -1.2 | 400 trees, depth 2 | no | no | – | tied with best (not a required model) |
| log10_width | ANN | -0.191 | 0.183 | -0.116 | -1.38 | 16 | yes | no | – | no better than constant |
| log10_width | GPR | 0.166 | 0.153 | 0.0017 | -1.39 | 5 | yes | no | – | tied with best |
| log10_width | TabPFN | – | – | – | – | – | – | – | – | not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download) |
| log10_width | ANN + physics | -0.0169 | 0.169 | 0.066 | -1.06 | 18 | yes | no | 0.0495 | worse than physics |
| log10_width | GPR + physics | 0.262 | 0.144 | 0.548 | -1.54 | 7 | yes | no | 0.00997 | chosen (selected ML model); not better than physics |

| target | label | P1_R2 | P1_RMSE | P2_R2 | P3_R2 | n_params | uncertainty | physically_consistent | perm_p | verdict |
|---|---|---|---|---|---|---|---|---|---|---|
| log10_width | TabPFN + physics | – | – | – | – | – | – | – | – | not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download) |
| log10_k | Constant | -0.149 | 0.387 | -0.201 | -2.34 | 1 | yes | yes | – | reference |
| log10_k | Physics | -0.149 | 0.387 | -0.201 | -2.34 | 1 | yes | yes | – | physics baseline (final model) |
| log10_k | Power law (OLS) | -0.306 | 0.412 | -0.178 | -6.25 | 4 | yes | no | – | no better than constant |
| log10_k | Ridge | -0.384 | 0.425 | -0.498 | -5.17 | 4 | no | yes | – | no better than constant |
| log10_k | SVR | -0.234 | 0.401 | 0.202 | -7.2 | 16 SV | no | no | – | no better than constant |
| log10_k | XGBoost | -0.493 | 0.441 | 0.0305 | -5.58 | 200 trees, depth 2 | no | no | – | no better than constant |
| log10_k | ANN | 0.408 | 0.278 | 0.365 | -4.19 | 21 | yes | no | – | tied with best |
| log10_k | GPR | -0.485 | 0.44 | -0.109 | -6.66 | 5 | yes | no | – | no better than constant |
| log10_k | TabPFN | – | – | – | – | – | – | – | – | not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download) |
| log10_k | ANN + physics | 0.399 | 0.28 | 0.386 | -4.47 | 22 | yes | no | 0.0198 | chosen (selected ML model) |
| log10_k | GPR + physics | -0.504 | 0.443 | -0.109 | -6.66 | 6 | yes | no | 0.621 | no better than constant |
| log10_k | TabPFN + physics | – | – | – | – | – | – | – | – | not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download) |

# C4 – Old pipeline vs final?

**Verdict:** Replace the old pipeline. COF: the old ANN on end-of-run COF scores $R^2$ +0.40 in P1 but +0.00 on the training runs (P2); the final model on COF_first10s scores +0.36 / +0.23 with two parameters. Wear: $R^2$ -0.48 / -0.47 (ANN on mean-of-locations k, plain levelling, replicate SD 0.34 dex) -> +0.29 / +0.15 (final model on the most-worn-location volume, replicate SD 0.07 dex).

| pipeline | quantity | target | model | n | R2 | RMSE | tercile_kappa | replicate_sd | signal_to_noise |
|---|---|---|---|---|---|---|---|---|---|
| old | COF | COF_end (old target) | ANN (plain) | 23 | 0.396 | 0.00933 | 0.358 | 0.00619 | 1.98 |
| old | wear | log10 k, mean of locations, plain levelling | ANN (plain) | 21 | -0.484 | 1.24 | -0.214 | 0.336 | 3.09 |
| final | COF | COF, first 10 s (primary) | physics | 23 | 0.36 | 0.00947 | 0.214 | 0.00557 | 2.17 |
| final | wear | log10 wear volume V [mm³] | physics | 16 | 0.289 | 0.387 | 0.0698 | 0.0668 | 7.1 |

# C5 – Does the literature data help?

**Verdict:** Do not use the literature data: paper COF ~ L^+0.52 v^-0.69 vs ours L^-0.09 v^-0.05; log10_COF: best with paper data GPR + study flag $R^2$ +0.24 vs ours alone +0.28; log10_k: best with paper data pooled + study offset: power law $R^2$ -0.31 vs ours alone +0.41. The published COF (0.002-0.007) and volumes (up to 1620 $mm^3$ in 200 s) are implausible as printed.

| target | strategy | uses_paper | n | R2_P1 | RMSE_P1 | bias |
|---|---|---|---|---|---|---|
| log10_COF | ours: power law | False | 23 | 0.231 | 0.033 | -0.00149 |
| log10_COF | ours: GPR | False | 23 | 0.0132 | 0.0374 | -0.00577 |
| log10_COF | ours: ANN | False | 23 | 0.278 | 0.032 | -0.00273 |
| log10_COF | naive pooling: power law | True | 23 | -20.8 | 0.176 | -0.0268 |
| log10_COF | naive pooling: GPR | True | 23 | -1.55 | 0.0601 | -0.00693 |
| log10_COF | naive pooling: ANN | True | 23 | -0.212 | 0.0414 | -0.00367 |
| log10_COF | pooled + study offset: power law | True | 23 | -3.28 | 0.0778 | -0.00146 |
| log10_COF | ours shrunk toward paper slopes | True | 23 | 0.231 | 0.033 | -0.00149 |
| log10_COF | paper slopes imposed | True | 23 | -53.3 | 0.277 | -7.35e-05 |
| log10_COF | GPR + study flag | True | 23 | 0.237 | 0.0329 | -0.00179 |
| log10_COF | ANN + study flag | True | 23 | -0.189 | 0.041 | -0.00242 |
| log10_k | ours: power law | False | 16 | -0.306 | 0.412 | 0.0215 |
| log10_k | ours: GPR | False | 16 | -0.485 | 0.44 | -0.00172 |
| log10_k | ours: ANN | False | 16 | 0.408 | 0.278 | 0.00455 |
| log10_k | naive pooling: power law | True | 16 | -4.1 | 0.815 | 0.149 |
| log10_k | naive pooling: GPR | True | 16 | -0.487 | 0.44 | 0.0154 |
| log10_k | naive pooling: ANN | True | 16 | -0.963 | 0.506 | -0.0178 |
| log10_k | pooled + study offset: power law | True | 16 | -0.315 | 0.414 | 0.0167 |
| log10_k | ours shrunk toward paper slopes | True | 16 | -0.488 | 0.44 | 0.0482 |
| log10_k | paper slopes imposed | True | 16 | -2.93 | 0.716 | 0.00617 |
| log10_k | GPR + study flag | True | 16 | -0.641 | 0.462 | -0.00524 |
| log10_k | ANN + study flag | True | 16 | -0.384 | 0.425 | -0.0229 |

| study | quantity | variable | exponent | ci_lo | ci_hi |
|---|---|---|---|---|---|
| ours | COF | load | -0.0859 | -0.123 | -0.0435 |
| ours | COF | speed | -0.0476 | -0.151 | 0.0462 |
| paper | COF | load | 0.525 | 0.0827 | 0.933 |

| study | quantity | variable | exponent | ci_lo | ci_hi |
|---|---|---|---|---|---|
| paper | COF | speed | -0.693 | -0.936 | -0.444 |
| ours | k | load | 0.0218 | -0.562 | 0.739 |
| ours | k | speed | -1.26 | -2.62 | 0.443 |
| paper | k | load | -1.45 | -2.1 | -0.848 |
| paper | k | speed | -0.334 | -0.678 | 0.0519 |

# C6 – Can the TA script be used?

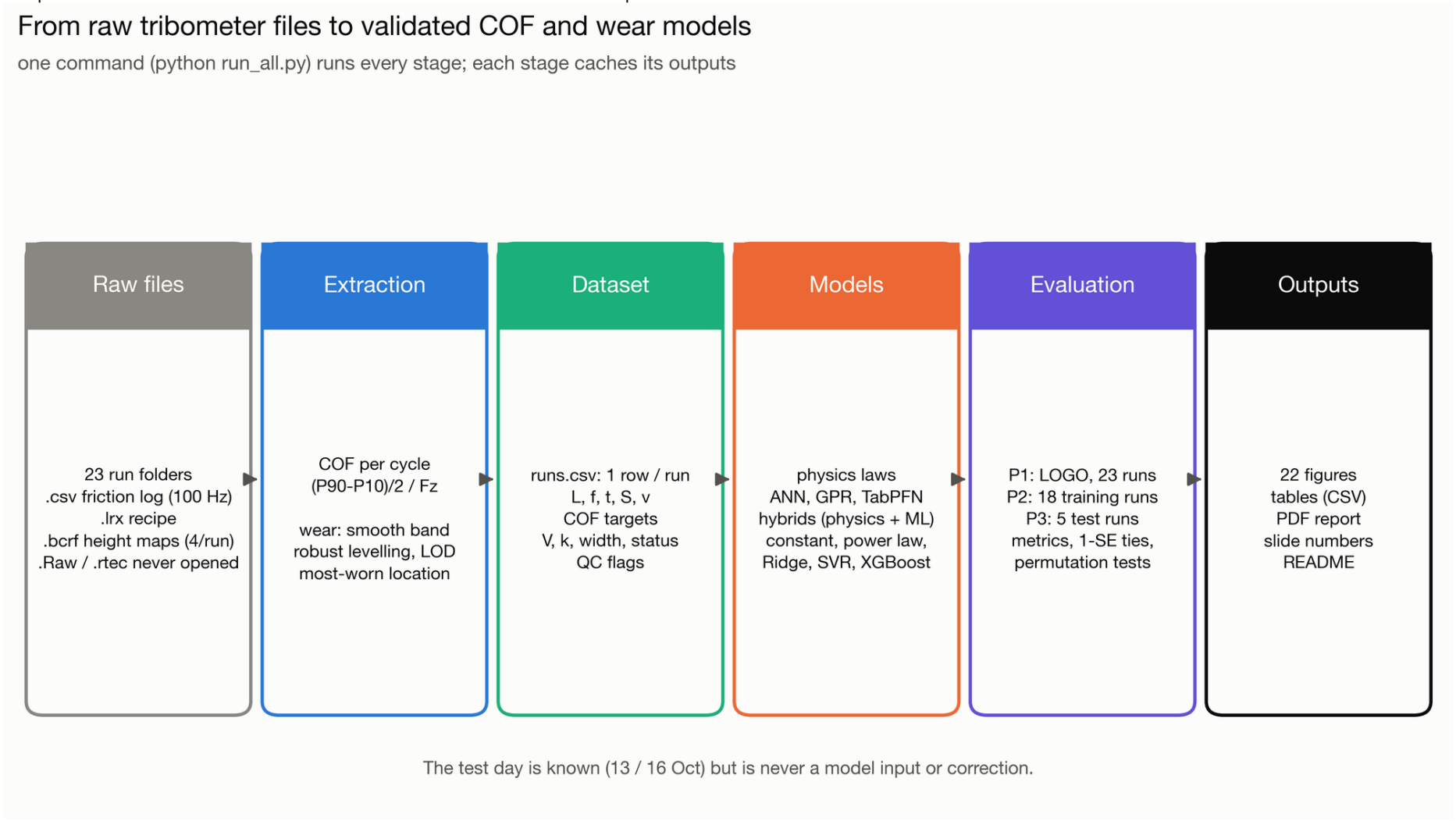
**Verdict:** Do not use the TA script as written: 25 of 92 scans are read along the track, and the R04/R14/R20 replicates spread 30x in k (SD 0.84 dex); fixed copy 0.13 dex, ours 0.07 dex (R14/R20 only). Archard $R^2$ (P1): original -0.03, fixed -0.19, ours +0.29.

| pipeline | k_R04 | k_R14 | k_R20 | replicate_sd_log10k | replicate_spread_factor | runs_with_volume | archard_R2_P1 | cof_law_R2_P1 |
|---|---|---|---|---|---|---|---|---|
| original TA script | 2.33e-06 | 7.02e-05 | 2.57e-06 | 0.841 | 30.1 | 21 | -0.028 | -0.138 |
| fixed TA copy | – | 5.62e-05 | 8.46e-05 | 0.126 | 1.51 | 20 | -0.192 | 0.246 |
| ours | – | 6.11e-05 | 4.92e-05 | 0.067 | 1.24 | 16 | 0.289 | 0.36 |

| bug | description | evidence |
|---|---|---|
| 1 | Orientation chosen from the unlevelled map (tilt decides) | 25 of 92 scans transposed, so the 'profile' runs along the track. R24.bcrf: transposed, reported depth 22.5 um (a pit), groove area 4352 um^2; with the correct axis the same code gives depth 7.4 um, groove area 0 um^2 |
| 2 | Points shallower than the cutoff zeroed one by one | 10 scans (correct axis) have groove area 0 although the depth is >= 3 um, e.g. R05b_13.bcrf (3.4 um), R11.bcrf (4.2 um), R13.bcrf (4.1 um), R14_a.bcrf (5.9 um) |
| 3 | Pile-up summed over the whole image outside the groove | median TA pile-up / band pile-up = 1.5 over 65 valid scans (texture peaks counted as pile-up) |
| 4 | Shoulder (reference) rows never checked to be unworn | in 23 of 90 scans with a track, the worn band reaches into the outer 15 % rows that the TA uses as the unworn baseline |
| 5 | Scan picked by the deepest point (texture valleys) | the deepest-point scan differs from the largest-groove valid scan in 12 of 20 runs with a detected groove |
| 6 | COF = mean of the instrument \|Fx\|/Fz channel; deceleration second not trimmed | TA COF / offset-free COF = 0.873 on average (13 % low); the ratio falls with frequency (r = -0.70, p = 0.000) |
| 7 | Hard-coded R01 parameters; friction log = first *.csv; outputs written into the data folder | R04 with the default arguments (10 N, 6 Hz, 120 s): k = 6.53e-06 instead of 2.33e-06 mm^3/(N m) (2.8x); run_pipeline() writes all_scans_breakdown.csv and single_scan_summary.csv into each run folder (avoided here) |

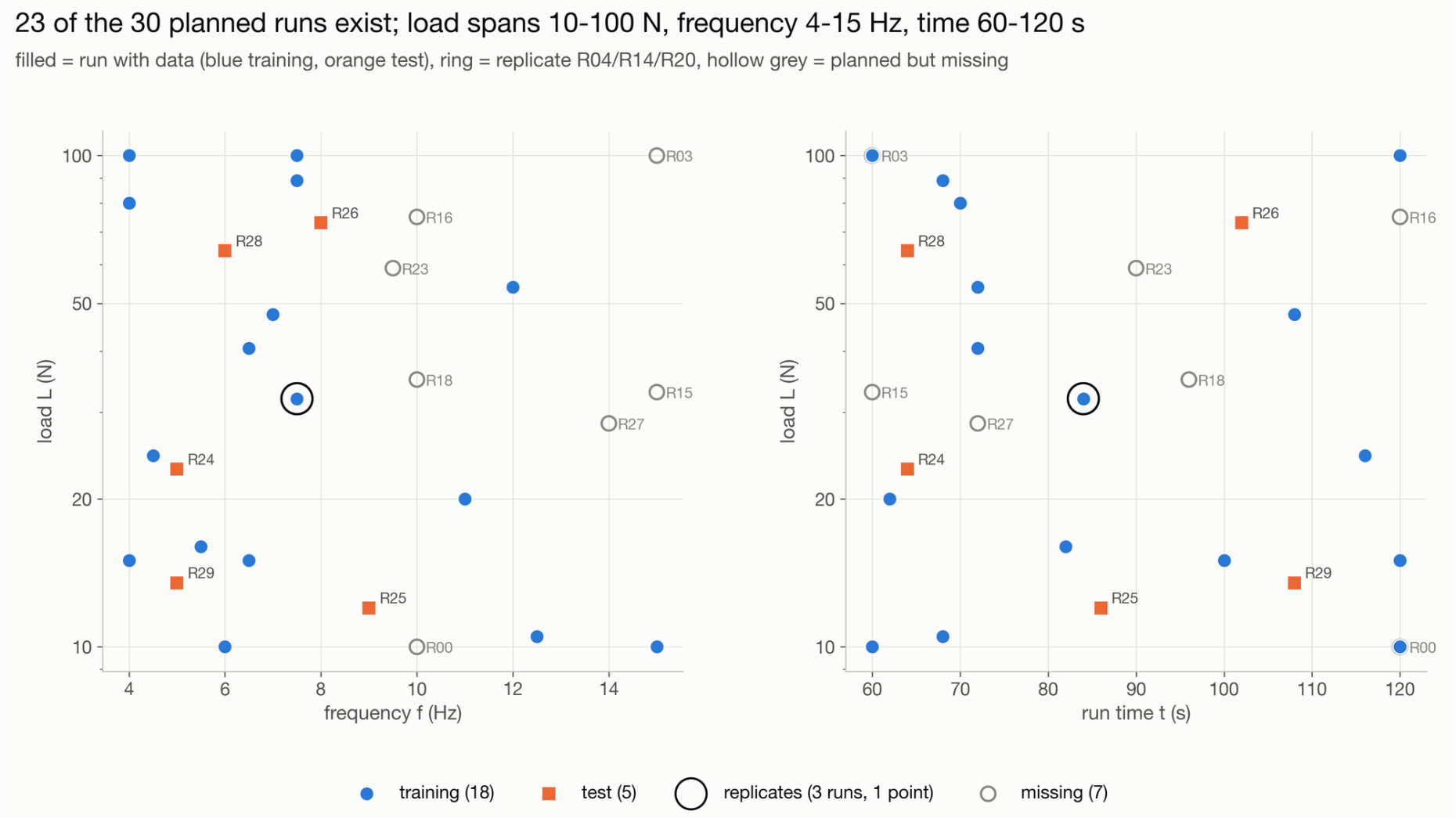
# Figures

Figure 1: Pipeline overview: raw files -> extraction -> dataset -> models -> evaluation -> outputs.



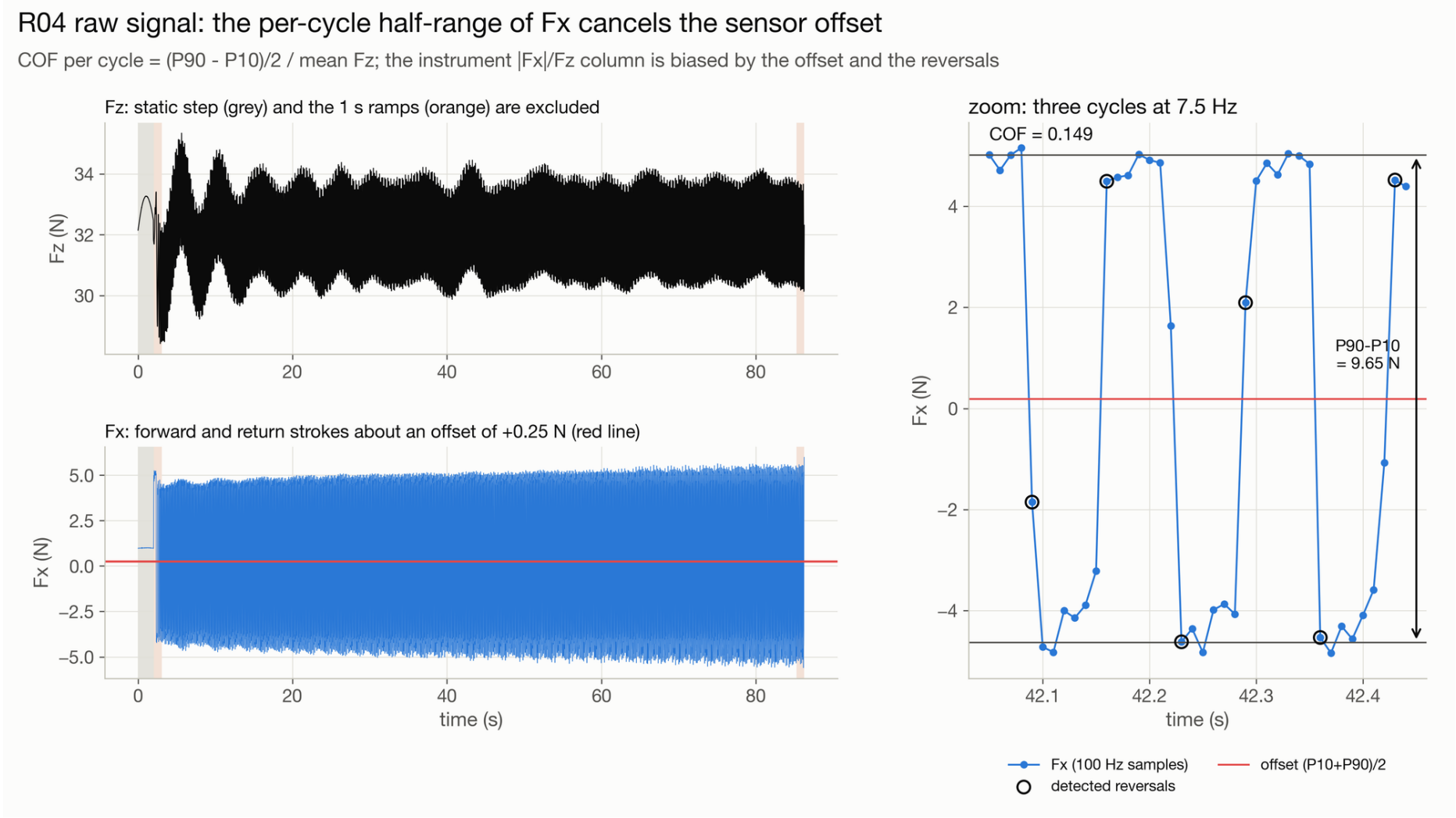

Data: code structure (src/tribo_final)

Figure 2: Test matrix of the 23 runs in L-f-t space; replicates ringed, missing DOE rows hollow.



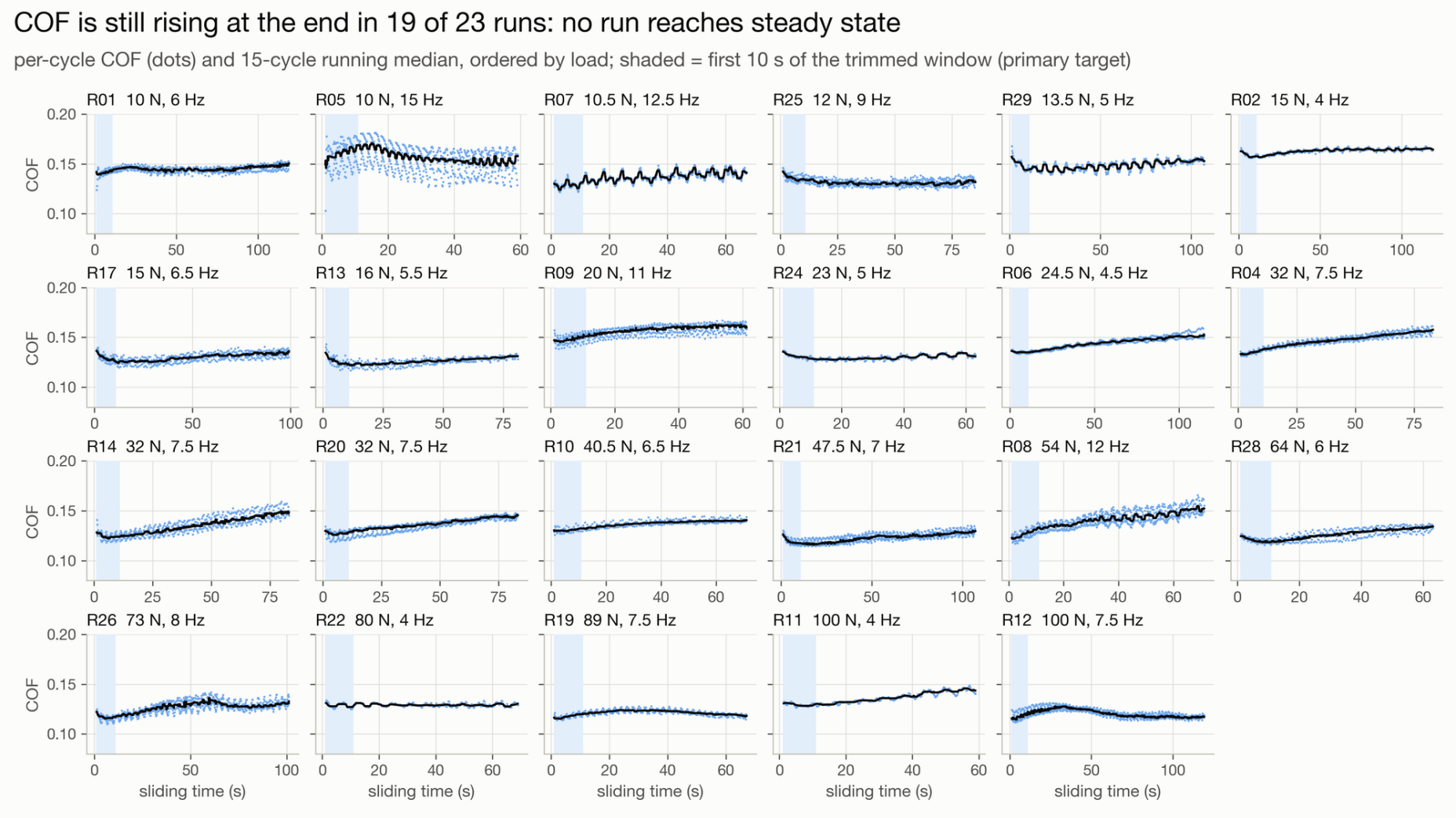

Data: outputs/data/runs.csv, inputs/run_sheet.csv

Figure 3: R04 raw Fz and Fx: static step, trimmed ramps, Fx offset, reversals and the per-cycle half-range COF.



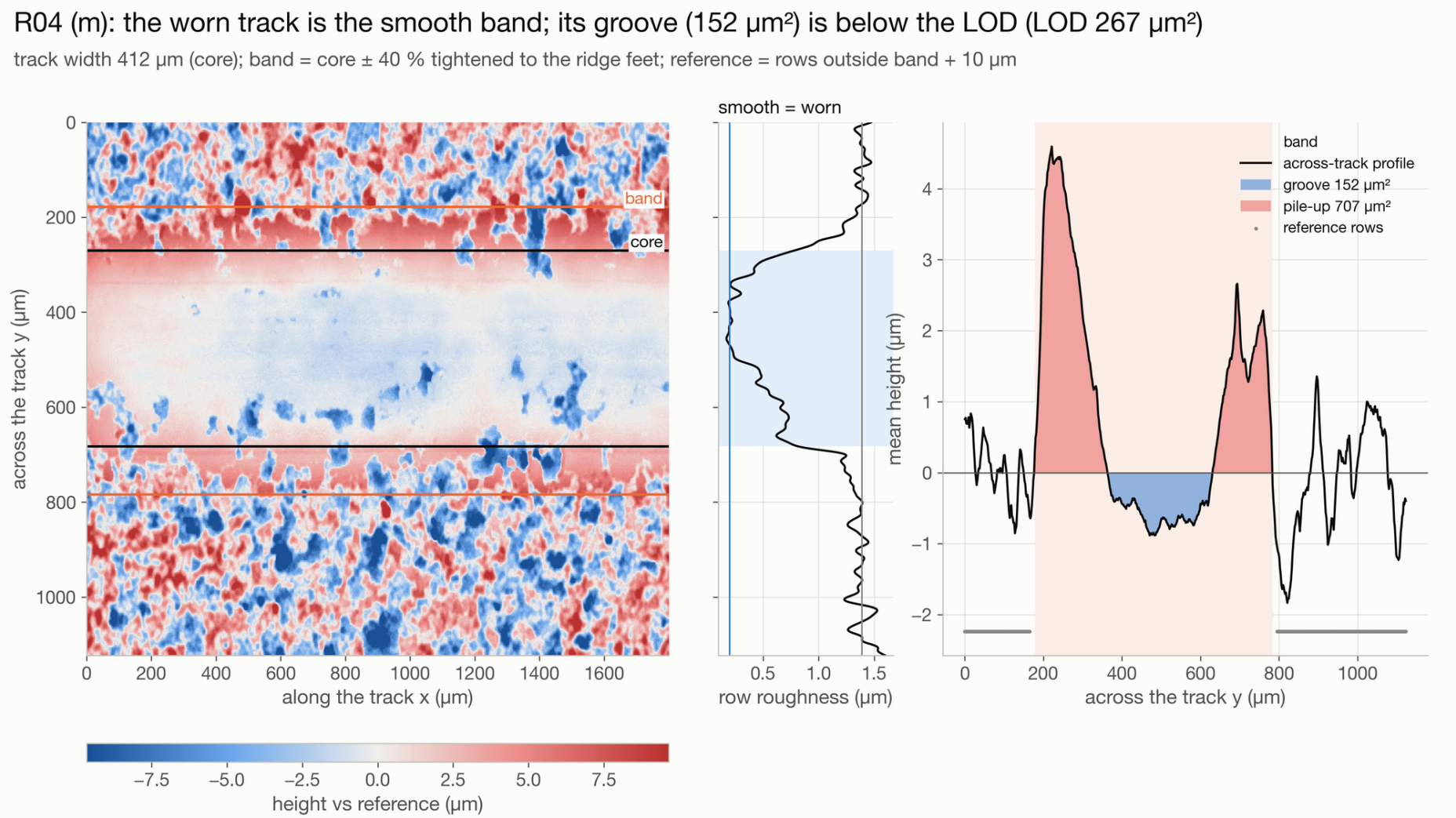

Data: testdata/R04/R04.csv

Figure 4: COF vs sliding time for all 23 runs (small multiples by load); first-10-s window shaded.



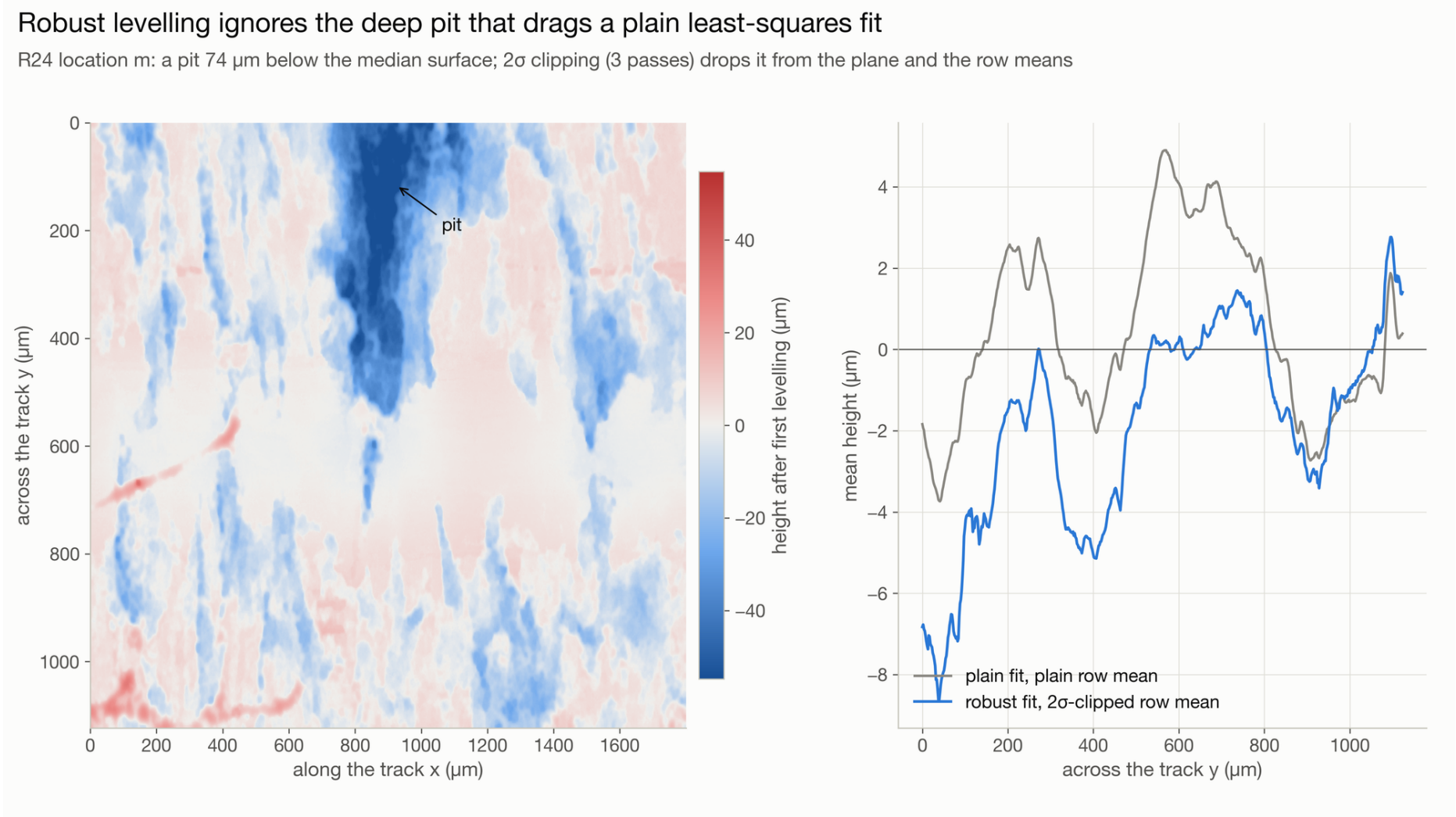

Data: outputs/data/per_cycle_cof.csv

Figure 5: Height map R04 (m) with core, band and reference rows; across-track profile with groove and pile-up shading.



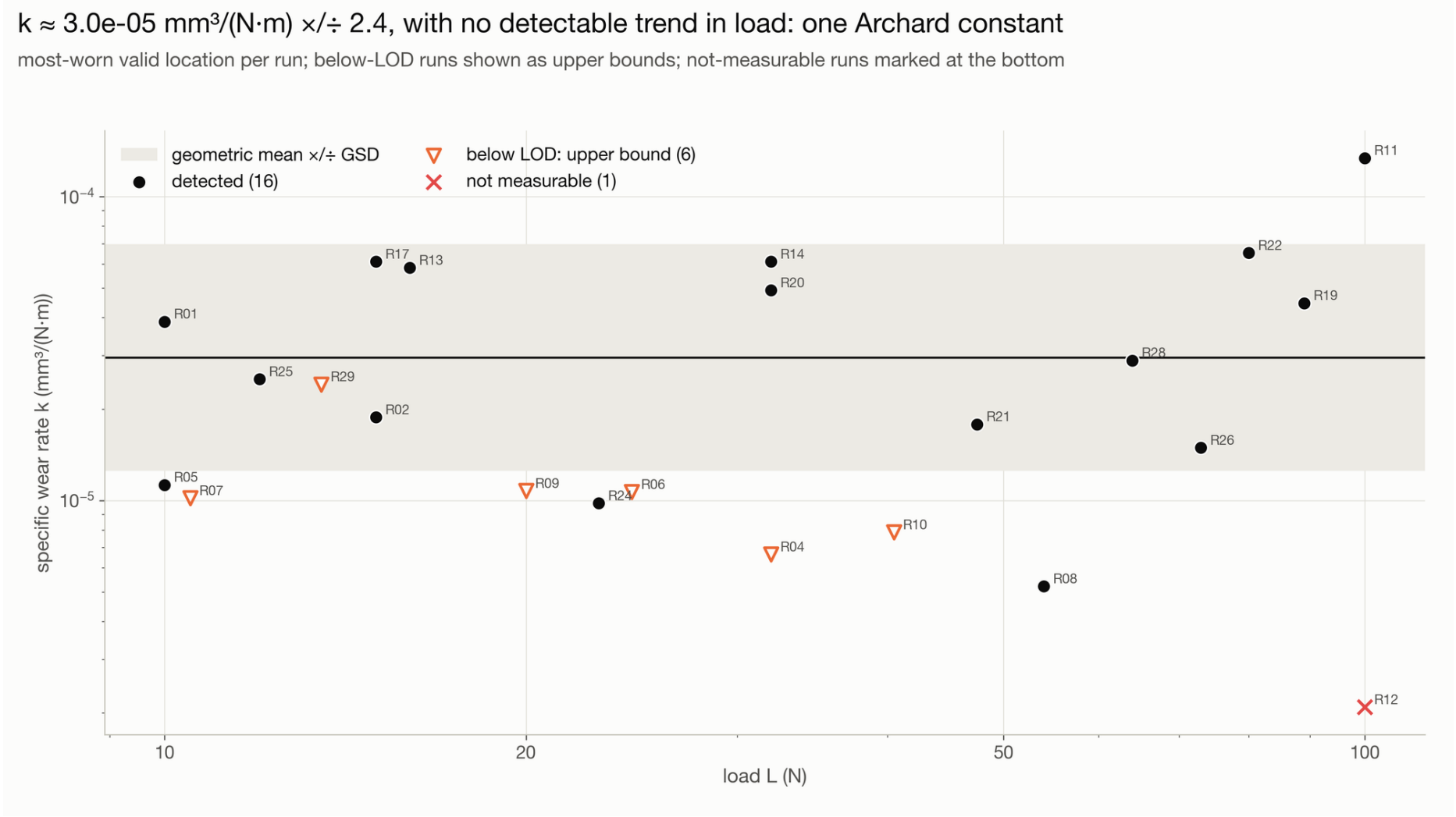

Data: testdata/R04/R04.bcrf

Figure 6: Robust vs plain levelling on R24 (m), a scan with a deep pit.



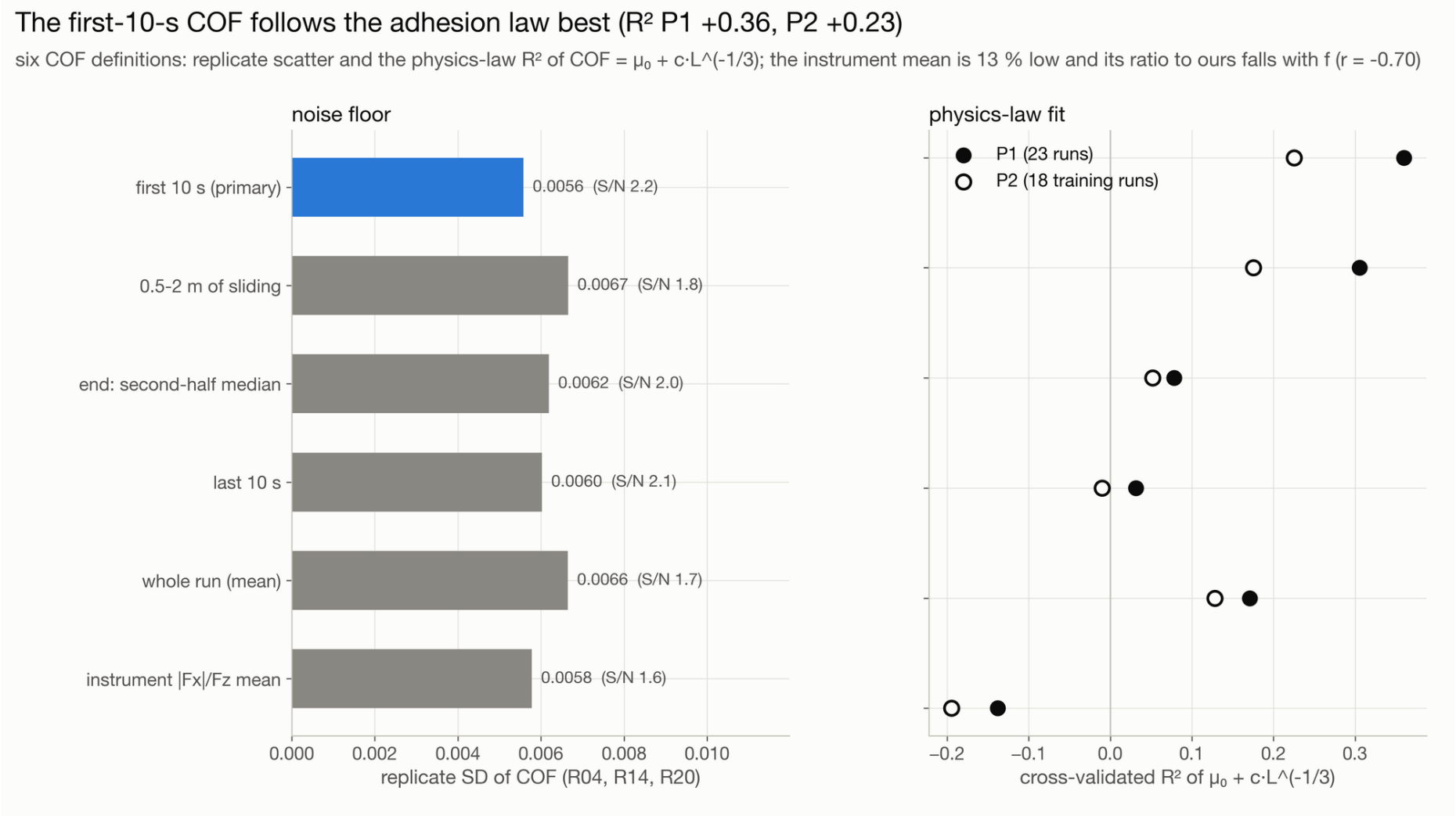

Data: testdata/R24/R24.bcrf

Figure 7: Specific wear rate k per run vs load with below-LOD upper bounds and not-measurable runs.



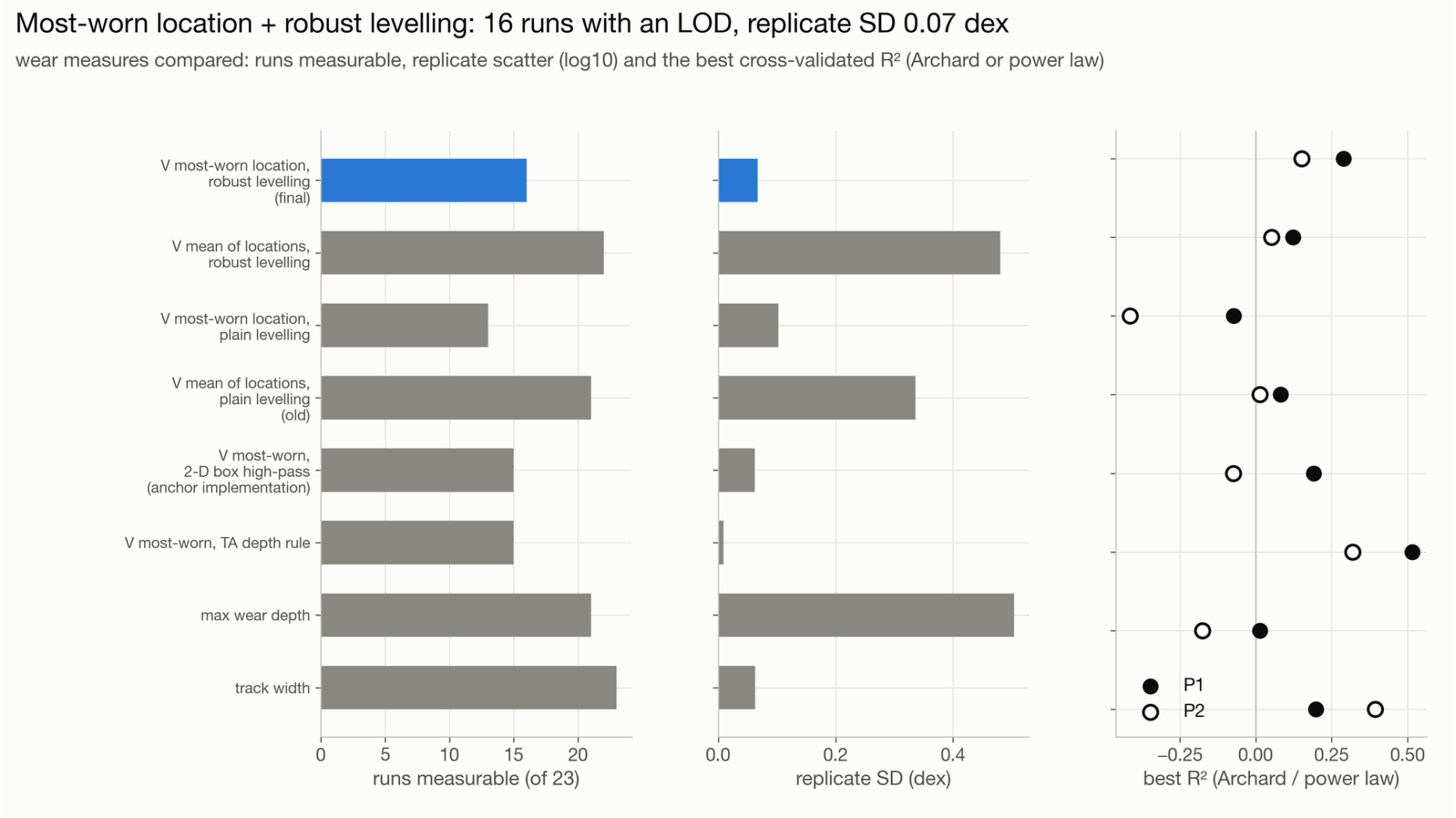

Data: outputs/data/runs.csv

Figure 8: C1: COF definitions compared by replicate SD and physics-law $R^2$ (P1, P2).



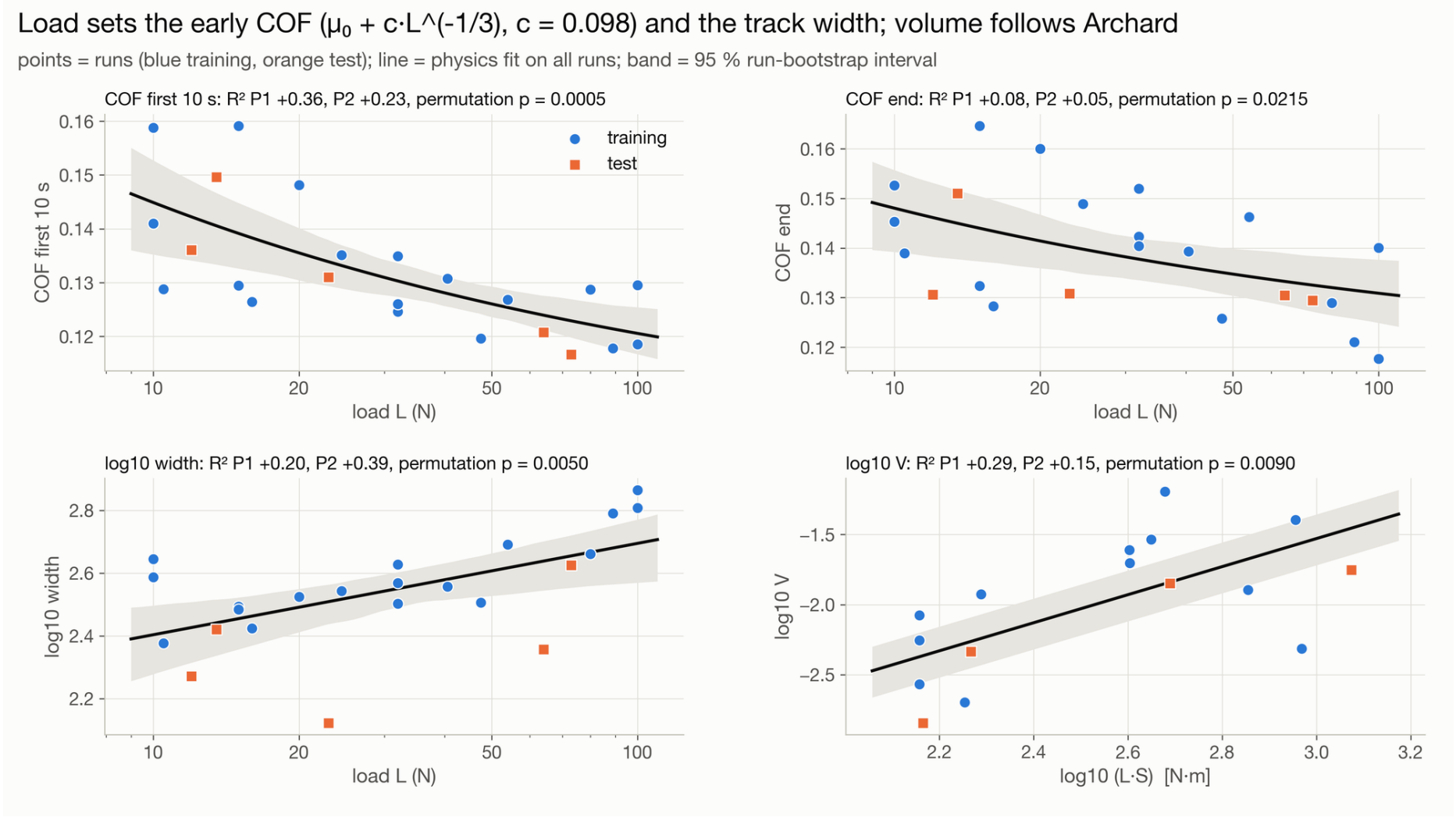

Data: outputs/tables/c1_cof_definitions.csv

Figure 9: C2: wear measures compared (runs measurable, replicate SD, best $R^2$ in P1/P2).



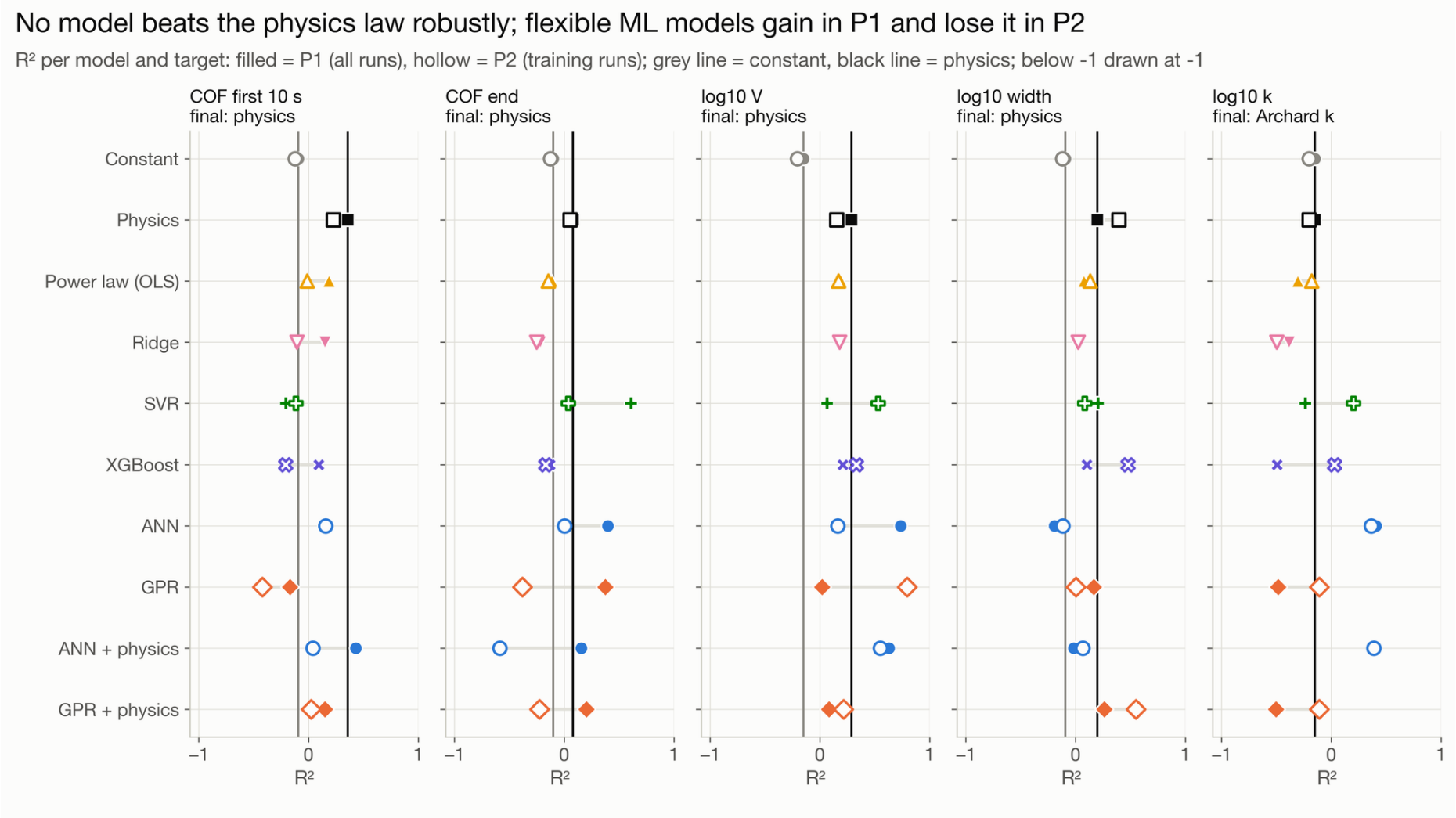

Data: outputs/tables/c2_wear_measures.csv

Figure 10: Physics fits: COF_first10s and COF_end vs load ($\mu_0$ + c·L^(-1/3)), track width vs load (log-log), wear volume vs L·S (Archard, slope 1), each with a 95 % run-bootstrap band.



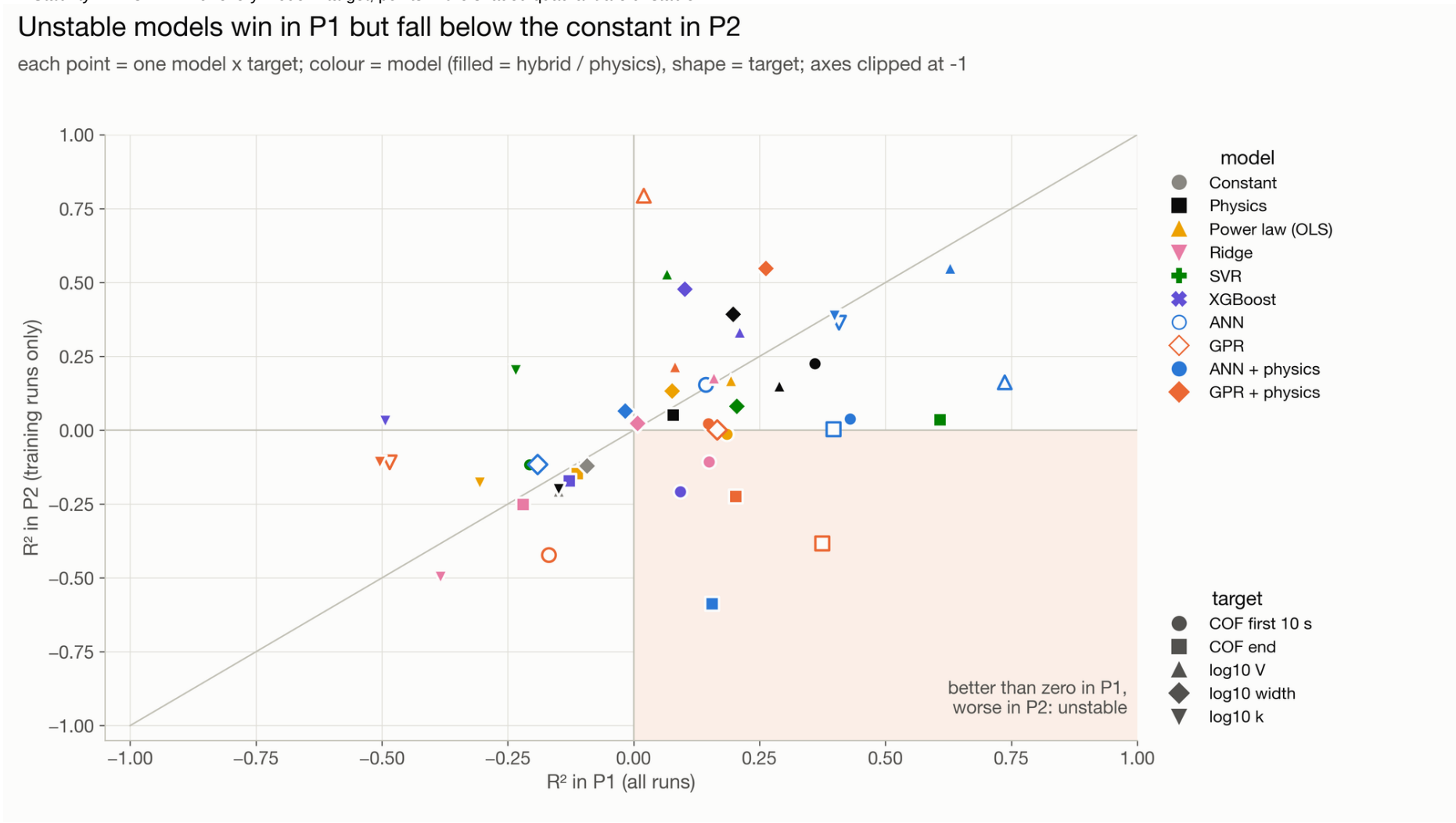

Data: outputs/data/runs.csv, outputs/tables/final_parameters.csv

Figure 11: C3: P1 (filled) and P2 (hollow) $R^2$ for every model and target, with constant and physics reference lines.



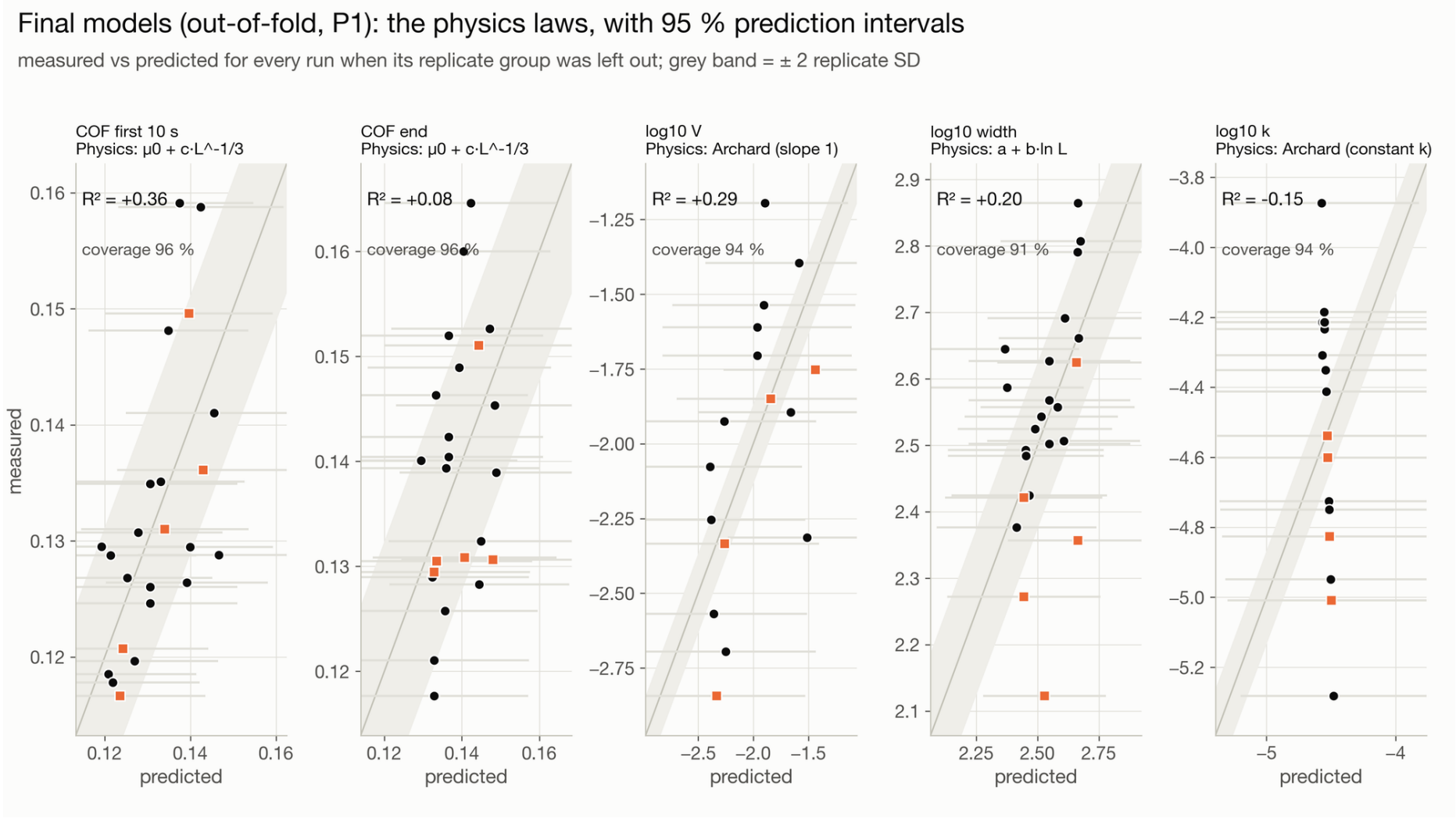

Data: outputs/tables/metrics_all.csv

Figure 12: Stability: P1 vs P2 $R^2$ for every model x target; points in the shaded quadrant are unstable.



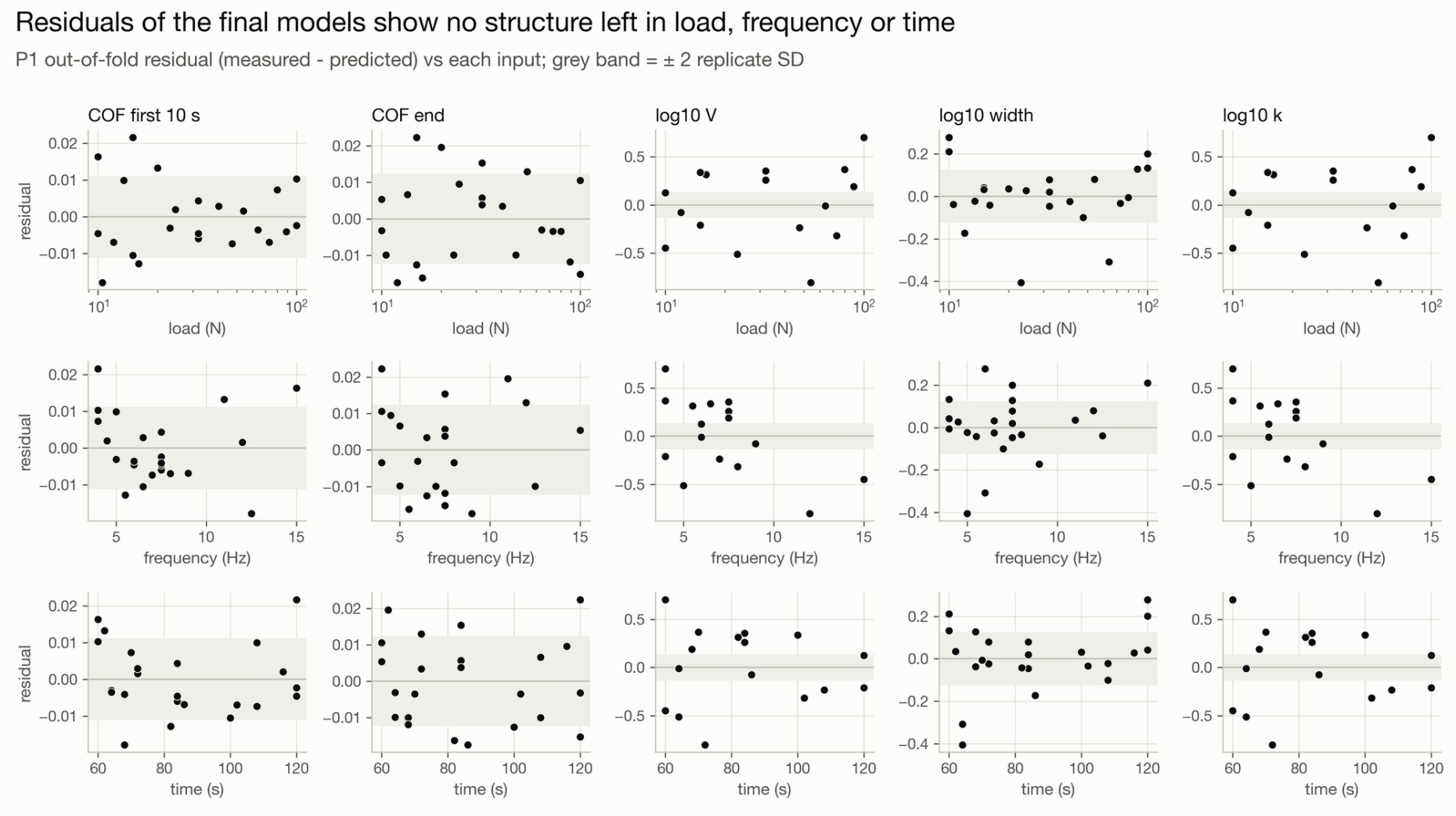

Data: outputs/tables/metrics_all.csv

Figure 13: Parity plots (P1 out-of-fold) of the final model per target with 95 % prediction intervals.



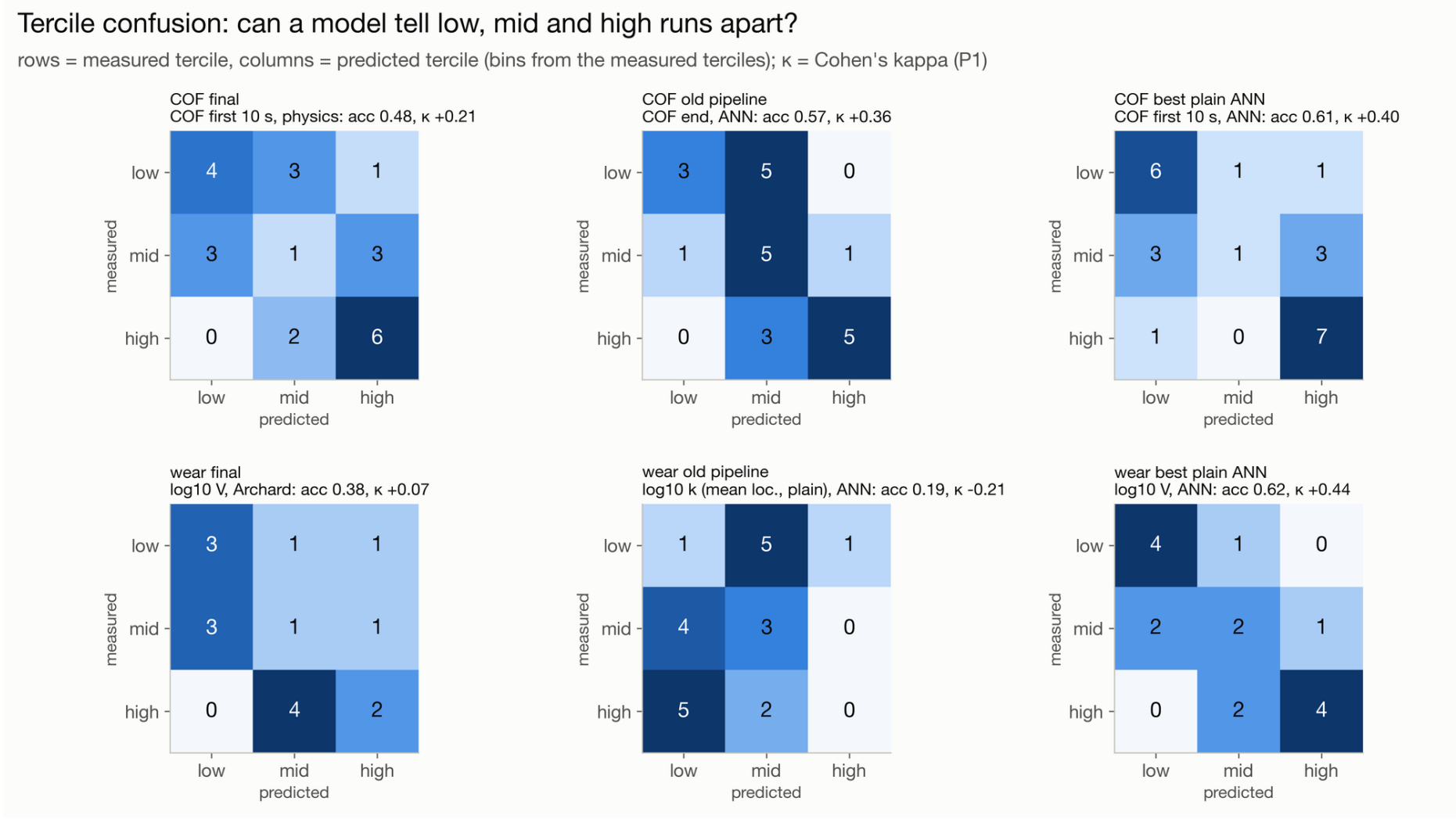

Data: outputs/data/oof_predictions.csv

Figure 14: Residuals of the final models vs load, frequency and time (P1 out-of-fold).



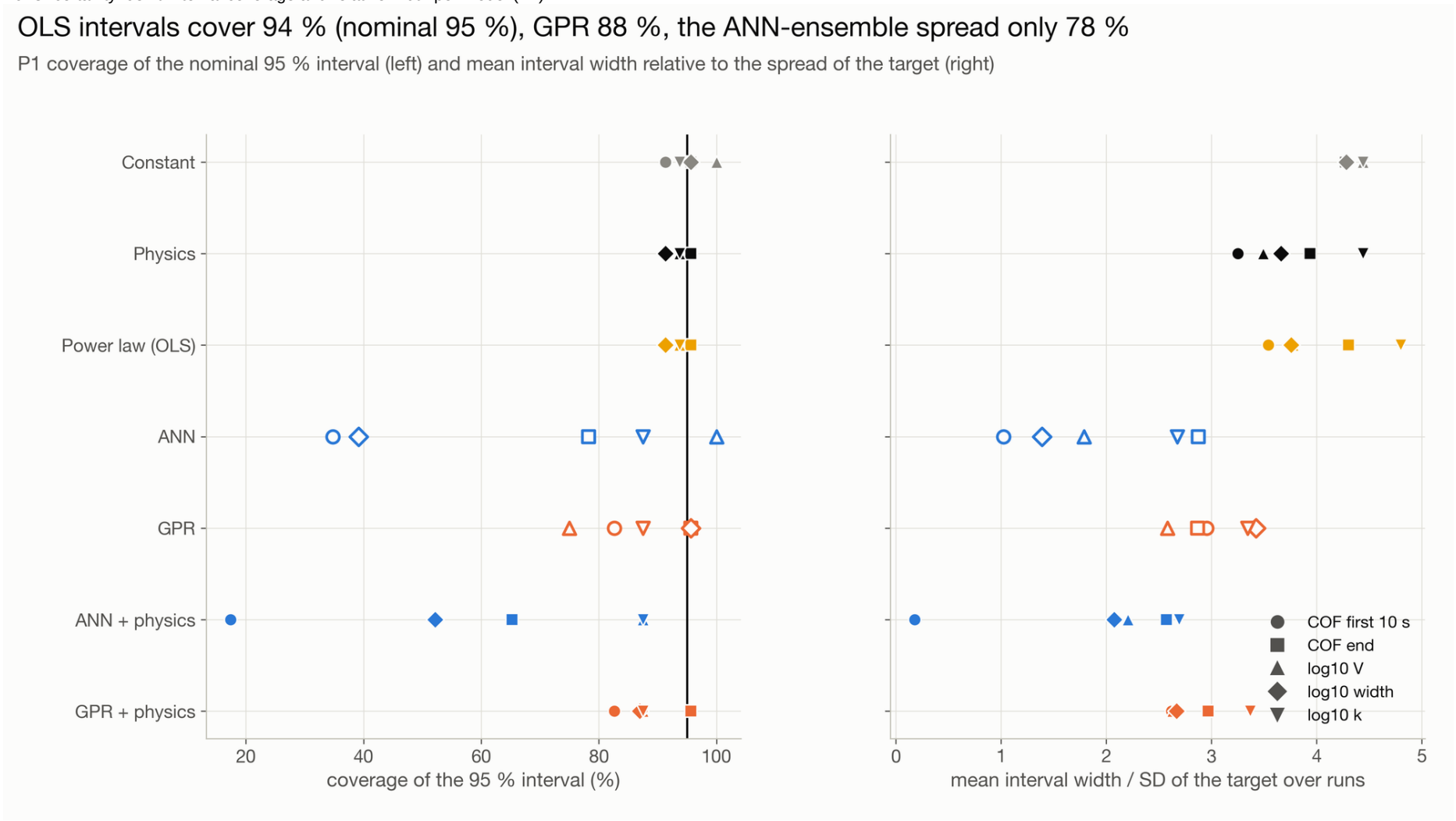

Data: outputs/data/oof_predictions.csv

Figure 15: Tercile confusion matrices (P1): final vs old pipeline vs best plain ANN, for COF and wear.



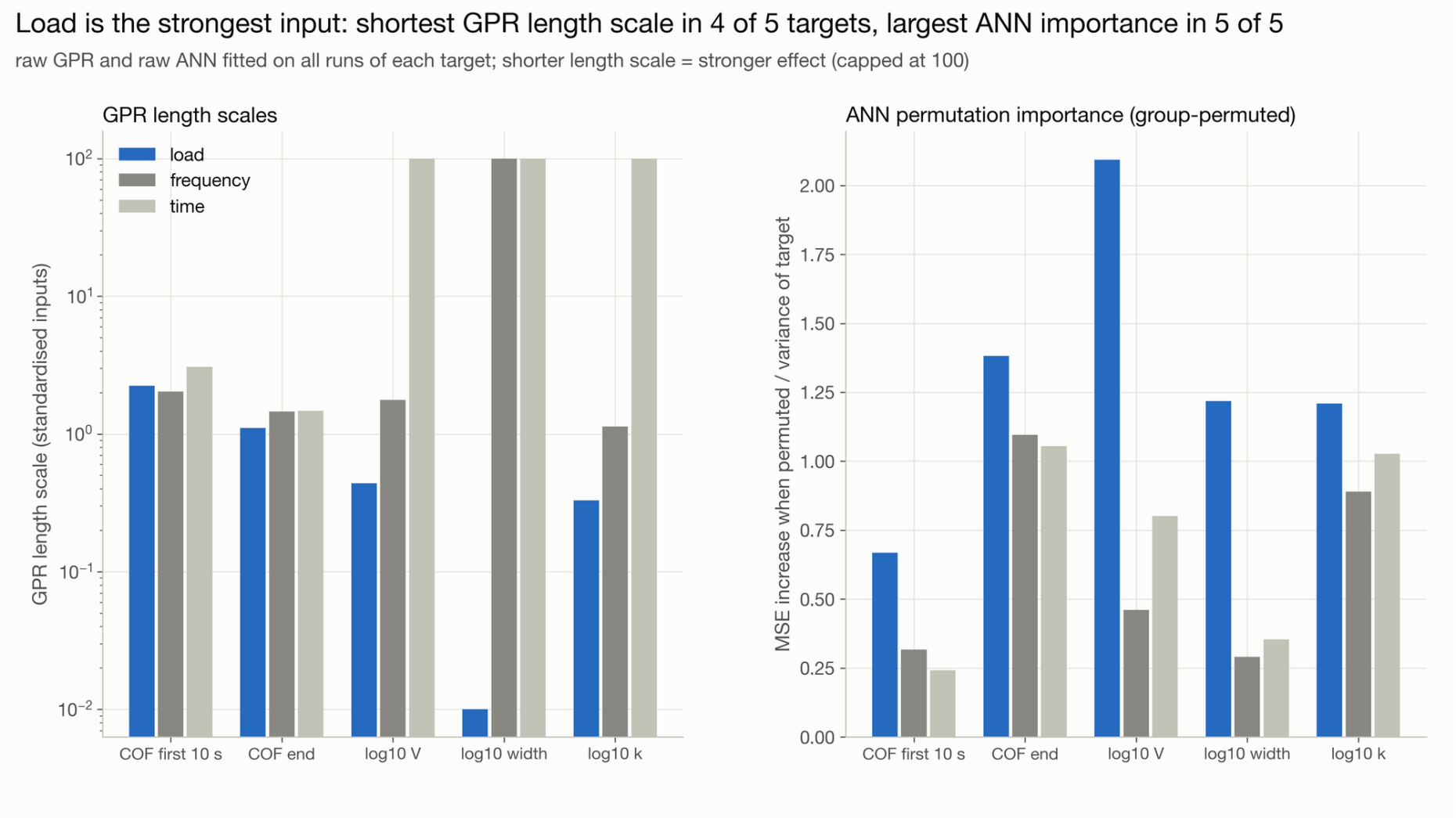

Data: outputs/data/oof_predictions.csv, outputs/data/c4_oof.csv

Figure 16: Uncertainty: 95 % interval coverage and relative width per model (P1).



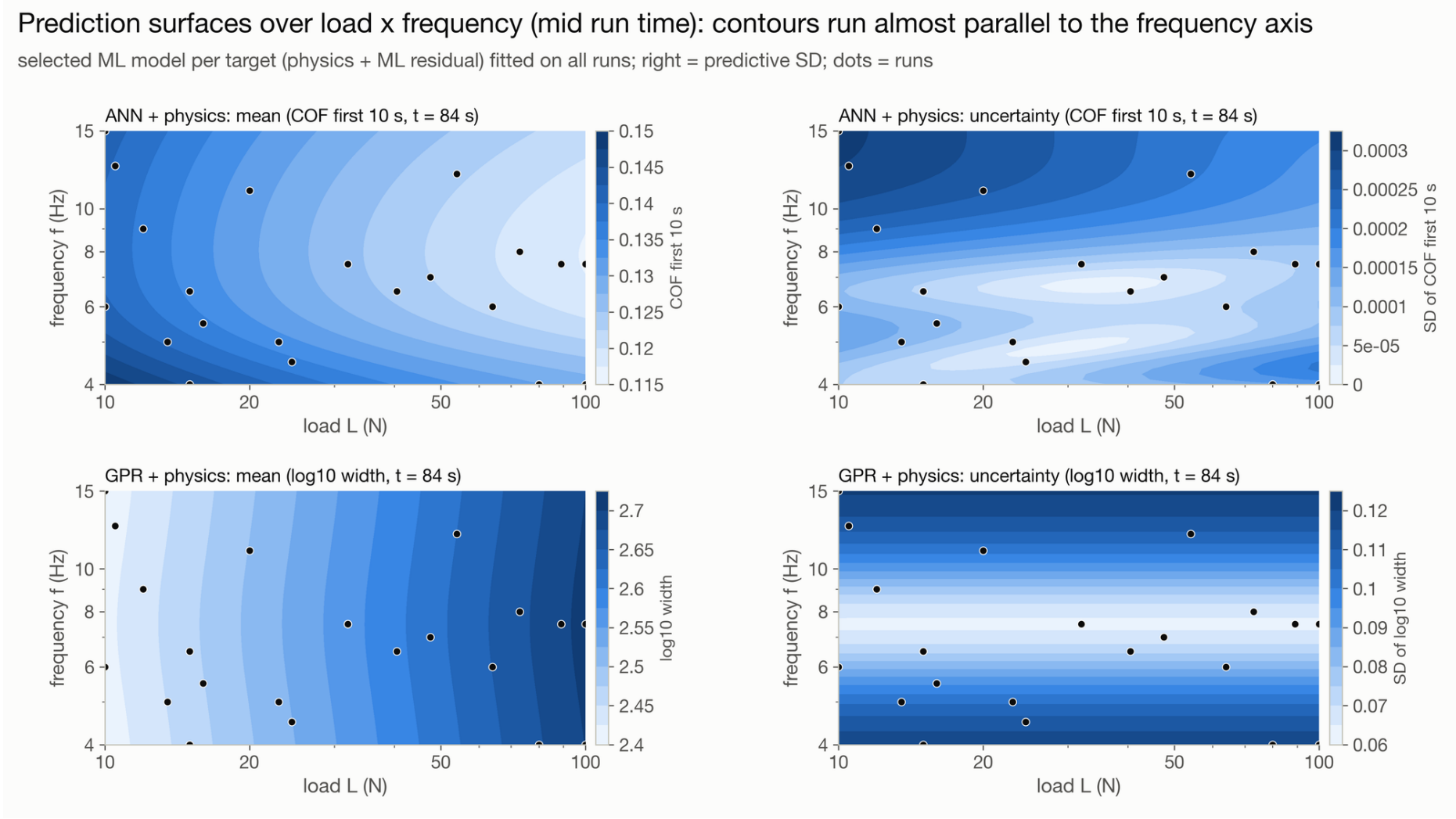

Data: outputs/tables/metrics_all.csv

Figure 17: Interpretation: GPR length scales and ANN permutation importance per input and target.



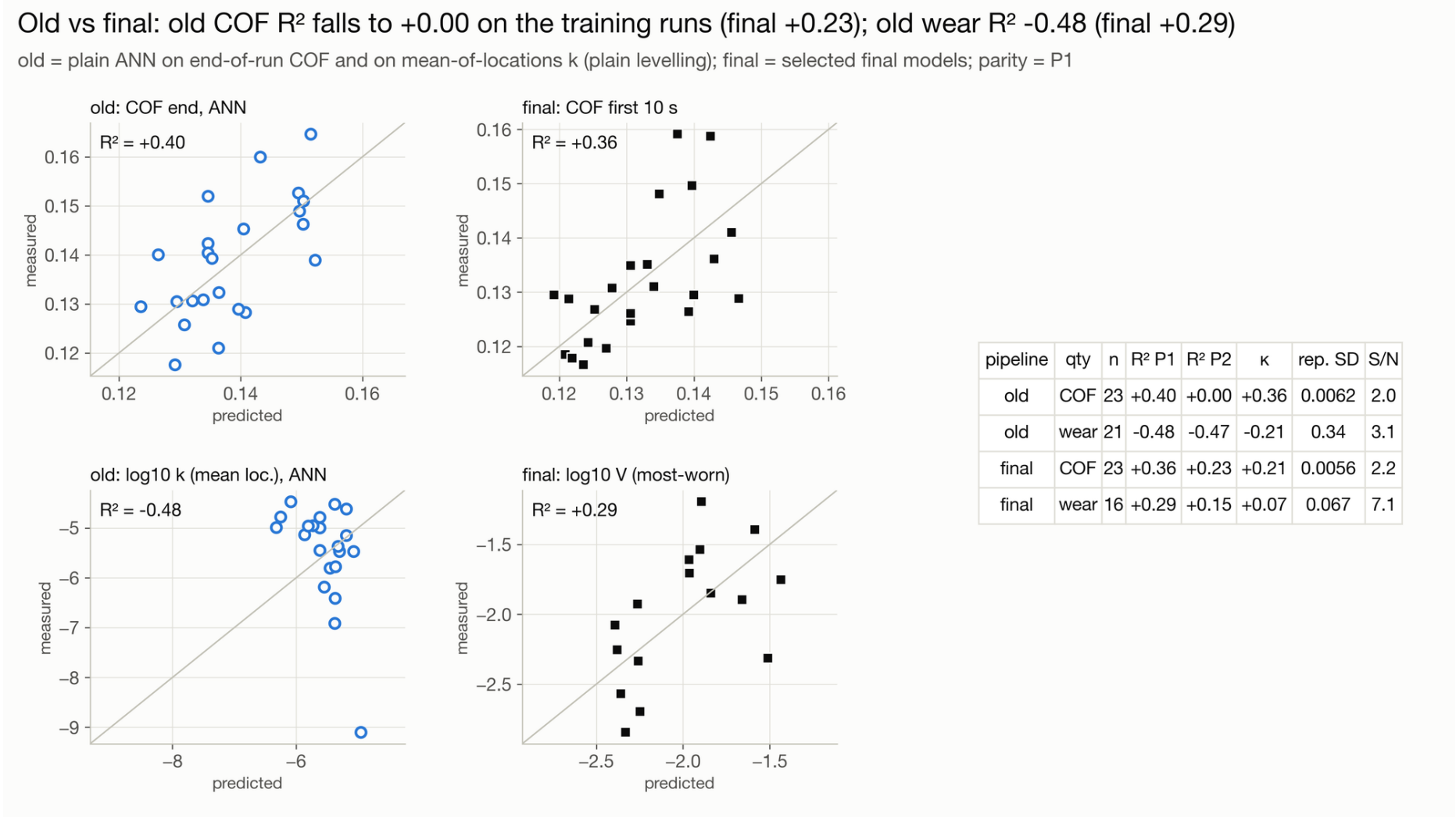

Data: outputs/tables/gpr_length_scales.csv, ann_permutation_importance.csv

Figure 18: Prediction surfaces (mean and SD) of the selected COF and width models over L x f at the median run time.




Data: outputs/data/runs.csv (models refitted on all runs)

Figure 19: C4: old pipeline vs final, metric table and P1 parity plots side by side.




| pipeline | qty | n | R² P1 | R² P2 | κ | rep. SD | S/N |
|---|---|---|---|---|---|---|---|
| old | COF | 23 | +0.40 | +0.00 | +0.36 | 0.0062 | 2.0 |
| old | wear | 21 | -0.48 | -0.47 | -0.21 | 0.34 | 3.1 |
| final | COF | 23 | +0.36 | +0.23 | +0.21 | 0.0056 | 2.2 |
| final | wear | 16 | +0.29 | +0.15 | +0.07 | 0.067 | 7.1 |

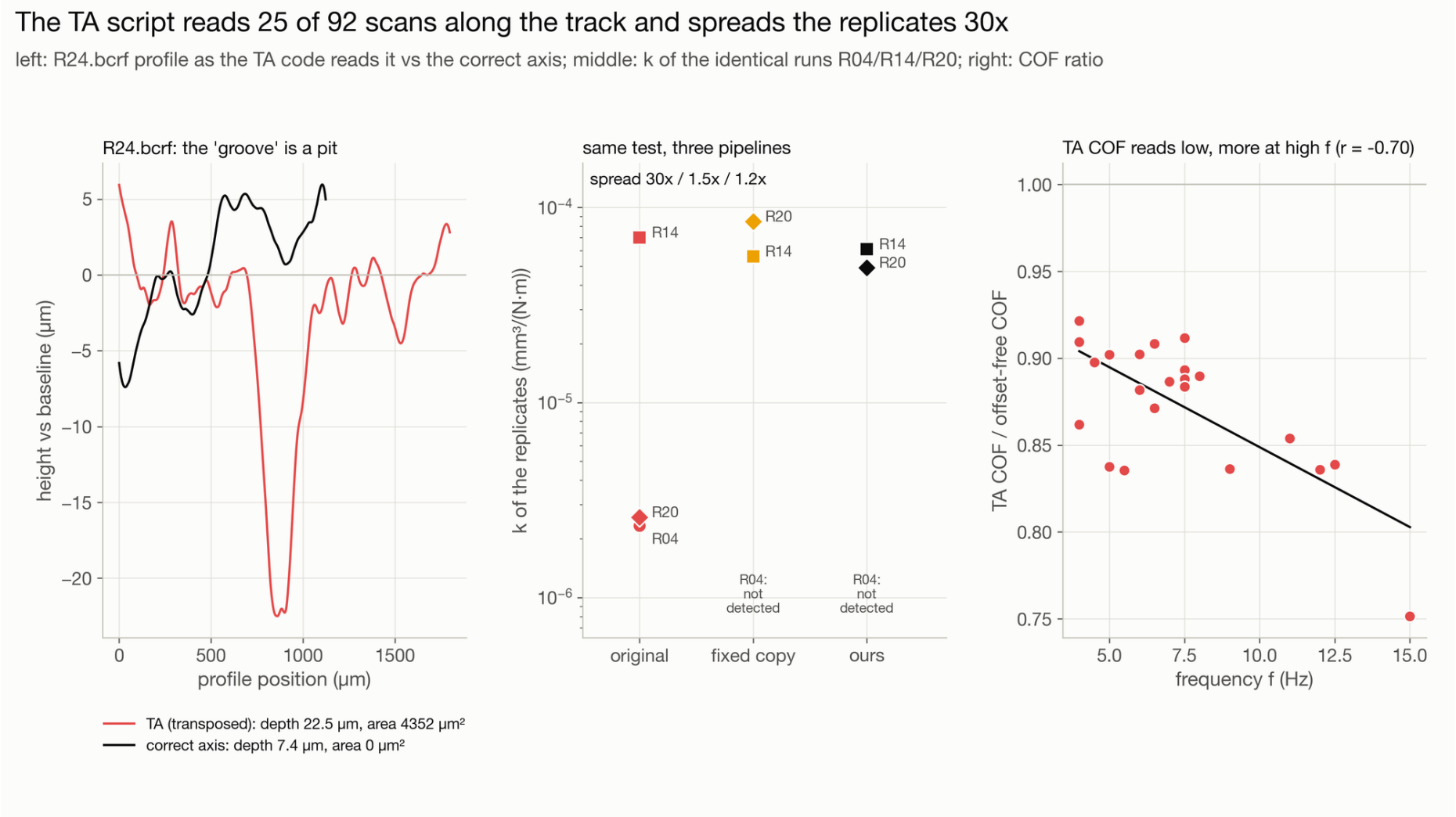
Data: outputs/tables/c4_old_vs_final.csv, outputs/data/c4_oof.csv

Figure 20: C5: relative COF and k vs load and speed, ours vs Ponnuswamy 2024, and the P1 scores of the pooling strategies.



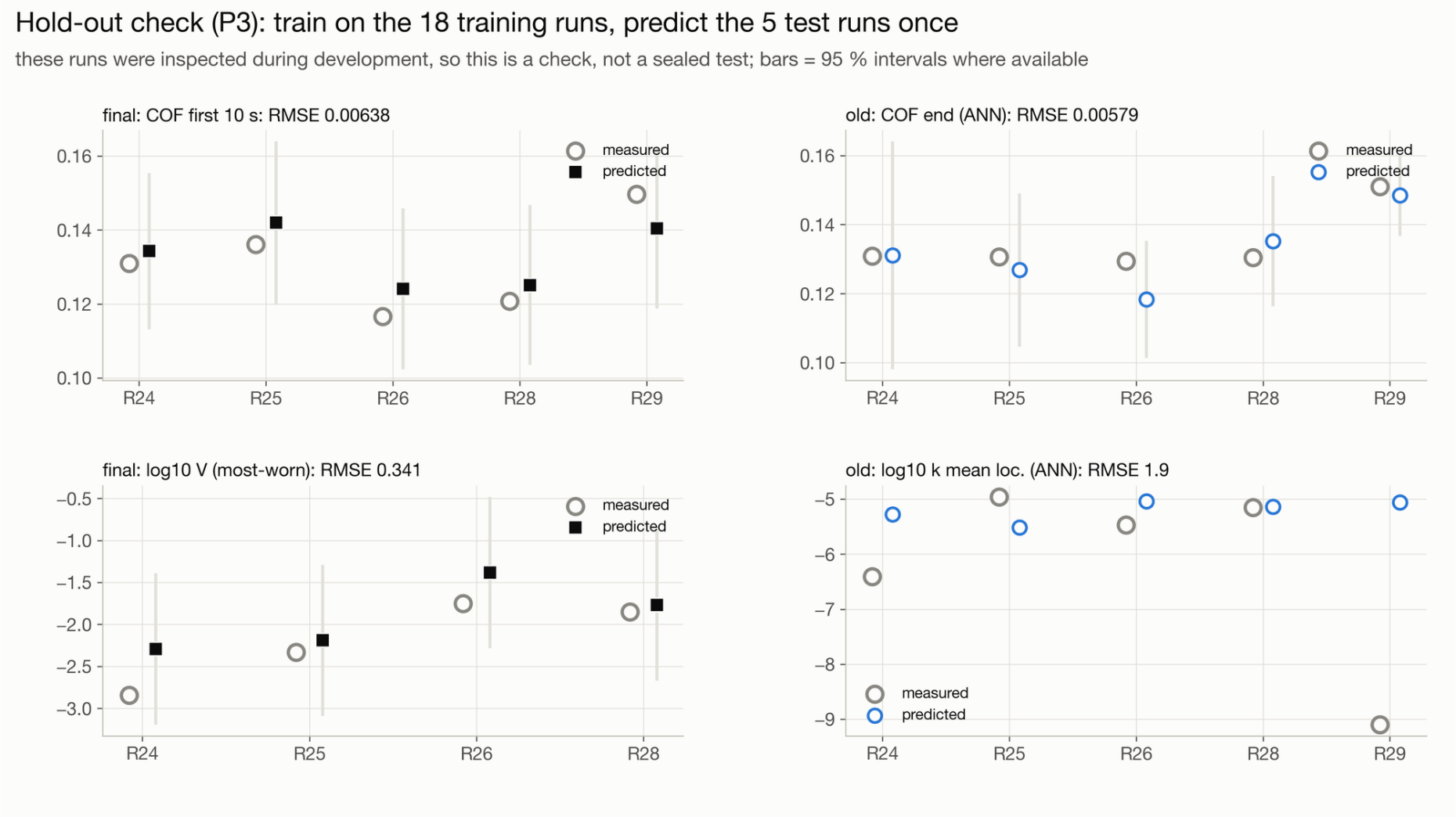

Data: inputs/ponnuswamy2024_table5.csv, outputs/tables/c5_trends.csv, c5_strategies.csv

Figure 21: C6: TA script - R24 orientation flip, replicate spread original/fixed/ours, COF ratio vs frequency.




Data: inputs/analyze_wear.py on testdata/, outputs/tables/c6_ta_*.csv

Figure 22: P3 hold-out: measured vs predicted for the 5 test runs, final vs old pipeline.




Data: outputs/data/c4_oof.csv